# GLFRNet: Global–Local Alignment and Frequency Reconstruction for Multimodal Remote Sensing Semantic Segmentation

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***Abstract*—High-precision semantic segmentation of multimodal remote sensing imagery (RSI) is critical for applications such as urban planning and land monitoring. However, existing methods often suffer from cross-modal semantic misalignment, geometric discontinuity, and loss of frequency information, resulting in blurred boundaries and incomplete object representations. To overcome these challenges, we present GLFRNet, a novel global–local alignment and frequency reconstruction fusion network for multimodal semantic segmentation. GLFRNet introduces three core innovations. First, a dual-stage structure-aware encoder (DSAE) is designed to combine convolutional and state space modeling, thereby enhancing both local texture extraction and global geometric feature representation. Second, a global–local feature projection alignment adapter (FPAA) is incorporated to explicitly align semantic and structural information between optical and digital surface model (DSM) modalities. Third, a phase-guided frequency reconstruction module (PGFRM) leverages the Fourier transform to jointly reconstruct low-frequency (LF) structures from DSM and high-frequency (HF) textures from optical images. Extensive experiments on the ISPRS Vaihingen and Potsdam datasets demonstrate that GLFRNet achieves state-of-the-art performance, with mean intersection-over-union (mIoU), overall accuracy (OA), and mean $F1$-score (m$F1$) scores of 87.03%, 91.93%, and 92.83% on Potsdam, and 85.35%, 93.27%, and 92.33% on Vaihingen, respectively. GLFRNet significantly outperforms previous methods, particularly in segmenting elongated objects, occluded regions, and complex terrain. All codes will be available on https://github.com/darkseid-arch/GLFRNet**

***Index Terms*—Feature alignment, frequency reconstruction, multimodal interaction, remote sensing, semantic segmentation.**

## I. Introduction

SEMANTIC segmentation of remote sensing imagery (RSI) is a fundamental technique for automated surface information extraction, supporting geospatial intelligence and knowledge modeling [1]. It plays a vital role in land use monitoring, environmental assessment, urban planning, disaster response, and national security [2]. Recent advances in remote sensing platforms and multimodal sensors, such as optical, SAR, and LiDAR, enable cross-scale data fusion, greatly enhancing the interpretation of heterogeneous geospatial information [3]. However, RSI presents higher spatial resolution, more complex object distributions, and greater multimodal heterogeneity than natural images. While multimodal data fusion improves recognition performance, it also brings challenges, including semantic misalignment, dynamic range disparity, and noise coupling. The need to balance global semantic consistency with structural detail under multimodal conditions has become a central issue in RSI semantic segmentation [4], driving new theories and methods for advanced geospatial understanding.

Traditional RSI semantic segmentation methods relied on hand-crafted features such as HOG [5], SIFT [6], and color histograms [7], combined with shallow classifiers or clustering algorithms. While these approaches achieved reasonable performance on low- and medium-resolution images, they struggled with complex backgrounds, multiscale objects, and heterogeneous data as spatial, spectral, and temporal resolutions increased. The advent of deep learning, particularly convolutional neural networks (CNNs), brought end-to-end feature learning and significantly advanced the field. CNN-based methods [8] substantially improved local texture and structural modeling but remained limited by their receptive fields, leading to insufficient global semantic reasoning and blurred object boundaries. More recently, transformers [9] and Mamba [10] have provided powerful context modeling and improved computational efficiency, further boosting segmentation performance. Nevertheless, most single-modal methods are primarily based on optical features and cannot fully leverage complementary information from other modalities.

With the increasing availability of multimodal remote sensing data, such as optical imagery, digital surface model (DSM), and multispectral sources, cross-modal fusion has emerged as a key research direction. Early fusion strategies based on CNNs, such as direct channel concatenation [11] or dual-branch architectures [12], have difficulty addressing modality-specific noise and distribution discrepancies. Recent works have introduced gating mechanisms [13], attention mechanism [14], and transformer-based frameworks [15] to enhance multimodal feature alignment and fusion. However, many existing approaches still adopt a single structure for feature fusion, resulting in an imbalance between local detail and global semantic consistency. Developing unified multimodal deep architectures that effectively integrate local and

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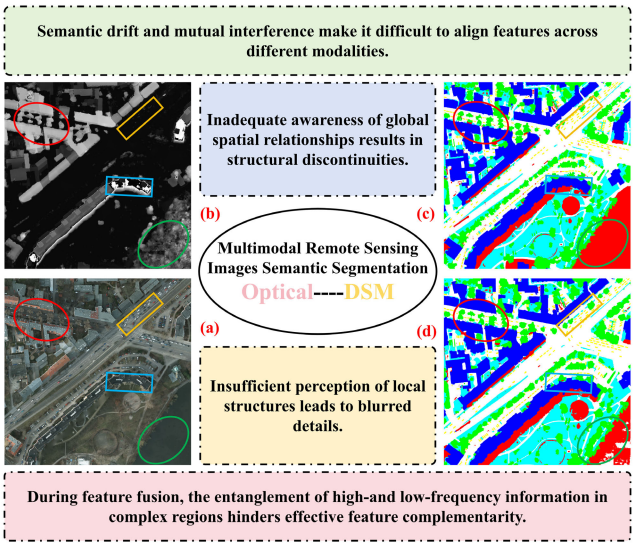



Fig. 1. Illustration of challenges in multimodal remote sensing semantic segmentation on optical-DSM data. (a) Optical image. (b) DSM image. (c) Ground truth. (d) Representative segmentation result with typical deficiencies.

global information, therefore, remains a major challenge in remote sensing semantic segmentation.

Despite the advances brought by multimodal deep learning, several key challenges remain for RSI semantic segmentation with complex structures and heterogeneous modalities, as illustrated in Fig. 1. Specifically: 1) current methods often lack unified modeling of global–local geometric structures, most focus on semantic-level fusion but fail to capture topological continuity, morphological geometry, and cross-scale relationships, resulting in structural discontinuities and blurred boundaries; 2) insufficient cross-modal alignment leads to semantic drift and mutual interference, as simple feature concatenation or attention-based fusion cannot fully resolve inherent differences in imaging mechanisms, spatial resolution, and noise among modalities, causing boundary mismatches and feature inconsistencies; and 3) most fusion strategies neglect frequency-domain modeling, overlooking the complementary properties of high-frequency (HF) (edges and textures) and low-frequency (LF) (global semantics) information, which limits cross-modal consistency and fine-grained detail recovery. These issues jointly restrict the accuracy and robustness of multimodal semantic segmentation in complex remote sensing scenarios.

To overcome the challenges of limited global–local structural perception, insufficient cross-modal alignment, and frequency-domain information loss in multimodal remote sensing semantic segmentation, we propose GLFRNet, a unified framework centered on structure-guided modeling, semantic alignment, and frequency-domain reconstruction. Specifically, GLFRNet comprises: 1) a dual-stage structure-aware encoder (DSAE) that synergistically captures global semantic context and local geometric details via lightweight convolutions and global state space modeling; 2) a feature projection alignment adapter (FPAA) that achieves robust cross-modal feature alignment through global projection and local adaptation; and 3) a phase-guided frequency reconstruction module (PGFRM) that leverages phase information in the frequency domain to enhance edge and texture preservation during fusion. This unified design enables accurate segmentation of complex objects with improved semantic consistency and structural fidelity in challenging multimodal scenarios. The main contributions of this article are summarized as follows.

1) We propose the DSAE, which synergistically integrates lightweight convolution and global state space modeling to jointly capture local geometric details and long-range semantic dependencies. This enables precise structural representation of complex and irregular objects in multimodal remote sensing images.
2) We design the FPAA, introducing a two-level alignment scheme that first projects features into a unified global semantic space and then performs local fine-grained adaptation. This approach effectively addresses modality heterogeneity and significantly improves cross-modal semantic consistency and structural alignment.
3) We introduce the PGFRM, which explicitly leverages phase information in the frequency domain to guide the fusion of HF (texture and edge) and LF (structural) components from different modalities. This ensures both semantic integrity and fine-detail preservation in multimodal segmentation tasks.

The article is structured as follows. Section II provides an overview of the related works. Section III introduces the proposed GLFRNet framework and its associated components. The experimental results and discussion analysis are presented in Section IV. Finally, the conclusion is drawn in Section V.

## II. Related Works

### A. Single-Modal Remote Sensing Semantic Segmentation

Single-modal remote sensing semantic segmentation aims to extract semantic information from a single modality for pixel-wise recognition and delineation of surface objects. As a foundation for multimodal perception, this task has seen rapid progress with the advent of deep learning. FCN [16] pioneered end-to-end pixel-level prediction but often exhibited blurred boundaries and loss of spatial detail due to simplified decoders. U-Net [17] addressed these issues with a symmetric encoder–decoder and skip connections, enabling effective multilevel feature fusion and achieving notable success in both medical and remote sensing domains. DeepLabv3+ [18] combines atrous spatial pyramid pooling (ASPP) with an efficient decoder for multiscale feature extraction and boundary refinement; PSPNet [19] aggregates contextual information via pyramid pooling; ResUNet-a [11] integrates dilated convolutions and multitask loss for precise segmentation of buildings and small objects. LANet [20] employs attention modules to improve deep-shallow feature fusion, further boosting multiscale object recognition and spatial detail restoration. Despite these advances, the limited receptive fields of CNNs hinder long-range dependency modeling and global semantic understanding. To address this, transformer-based architectures

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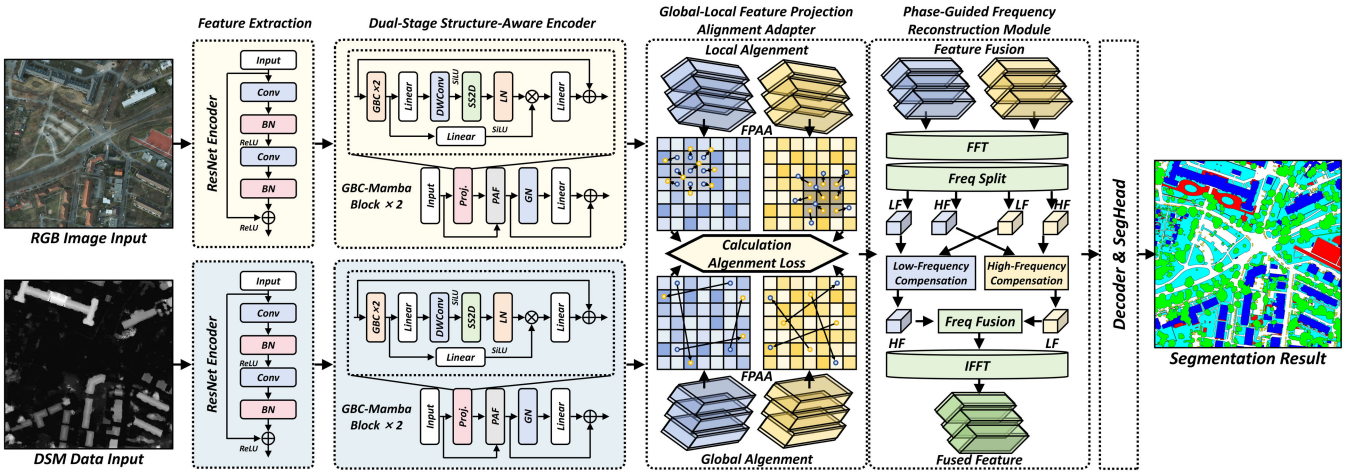



Fig. 2. Overview of the proposed GLFRNet architecture. The framework consists of parallel encoders for optical and DSM inputs, DSAE for feature extraction, global–local feature alignment via FPAA, PGFRM for cross-modal compensation, and a multiscale decoder for semantic segmentation.

[9] leverage self-attention for explicit global context modeling. Vision transformer (ViT) [21] initiated this paradigm, followed by SegFormer, Swin transformer, and Segmenter, which further improved global representation and boundary accuracy through hierarchical and multiscale designs. Hybrid models such as MMT [22], CTCFNet [23], and UNetFormer [24] integrate CNNs and transformer strengths for enhanced segmentation in complex urban scenes. Recently, state space models (SSMs), exemplified by Mamba [25], have emerged as efficient architectures for long-sequence modeling with linear complexity. Integrated frameworks such as SambaNet [26] and GLVMamba [27] validate the scalability of SSMs, while DiffMamba [28] improves small object and multiscale consistency via diffusion and phase-aware modules.

### B. Multimodal Remote Sensing Semantic Segmentation

Multimodal remote sensing semantic segmentation has emerged as a key direction for addressing the limitations of single-modal perception in complex scenes. By integrating heterogeneous data sources, such as optical, DSM, SAR, and depth, these methods significantly enhance object recognition accuracy, robustness, and scene understanding across spectral, spatial, and structural dimensions. Early approaches primarily focused on feature-level alignment and hierarchical fusion. TransUNet [29] unified the global dependency modeling of transformers with the localization ability of U-Net. CMFNet [30] and FuseNet [12] introduced cross-modal multiscale fusion and dual-branch encoders to explicitly leverage complementary information between modalities. ESANet [31] and SA-GATE [32] further improved real-time efficiency and noise suppression by lightweight design and gated cross-modal interactions. SkySense++ [33] proposes a semantic-enhanced multimodal foundation model for Earth observation with strong cross-modal generalization. With the development of SSMs, especially Mamba, multimodal segmentation saw notable advances in long-range dependency modeling and cross-modal consistency. DS-S4 [34] combines Mamba and diffusion models to enhance both semantic consistency and geometric structure preservation. MBSSNet [35] introduces selective scanning and cross-modal fusion to mitigate semantic drift between optical and SAR/DSM. HMAFNet [36] employs spatial-channel Mamba and information-guided fusion for joint global and local modeling. RS3Mamba [37] and MambaTriNet [38] leverage multibranch and completion modules for multiscale feature integration and structural restoration. Recent research has increasingly focused on high-resolution data, lightweight architectures, and cross-modal robustness. MEET [39] provides a large-scale benchmark for fine-grained scene classification from zoom-free RSI. CMGFNet [13] and MIEFNet [40] utilize dual encoders, gated fusion, and transformer-based interactions to extract fine-grained building and terrain features. MultiSenseSeg [41] and MFTransNet [42] propose modality expert networks and hybrid CNN-transformer designs for flexible feature extraction and efficient segmentation. FTransUNet [43] and FTransDeepLab [44] integrate convolutional and attention mechanisms for collaborative global–local feature modeling and precise boundary recovery. MFNet [45] and CFCGNet [46] enhance generalizability and structural consistency via adapter tuning, pyramid fusion, and graph-based compensation. The mKGR framework [47] introduces multigranularity knowledge graph reasoning for zero-shot urban land-use mapping based on semantic relationships. SACFNet [48] and CSFAFormer [49] further improve cross-modal alignment and computational efficiency by multidirectional fusion and class-adaptive calibration.

## III. Methodology

### A. Overall Architecture

The proposed GLFRNet framework is designed to fully exploit the complementary information between optical imagery ($\boldsymbol{I}_{\text{opt}} \in \mathbb{R}^{3\times H\times W}$) and DSM data ($\boldsymbol{I}_{\text{dsm}} \in \mathbb{R}^{1\times H\times W}$) for precise semantic segmentation in complex scenes. As illustrated in Fig. 2, the framework first employs two parallel encoders, $\boldsymbol{F}^{\text{opt}}_{\text{Encoder}}$ and $\boldsymbol{F}^{\text{dsm}}_{\text{Encoder}}$, to extract hierarchical features $\boldsymbol{F}^{\text{e}}_{\text{opt}}$ and $\boldsymbol{F}^{\text{e}}_{\text{dsm}}$ from the optical and DSM modalities, respectively. Each encoder is equipped with stacked DSAEs, enabling the network to model both global context and local

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spatial relationships for each modality. To address cross-modal semantic misalignment, a FPAA is introduced to project the features $\boldsymbol{F}_{\text{opt}}^{\text{e}}$ and $\boldsymbol{F}_{\text{dsm}}^{\text{e}}$ into a unified latent space, yielding aligned representations $\boldsymbol{F}_{\text{opt}}^{\text{align}}$ and $\boldsymbol{F}_{\text{dsm}}^{\text{align}}$. This module adaptively aligns global and local feature distributions, supported by a global–local alignment loss that enforces both holistic and region-wise consistency between the two modalities. Subsequently, GLFRNet incorporates a PGFRM to further exploit cross-modal complementarity. In this stage, $\boldsymbol{F}_{\text{opt}}^{\text{align}}$ and $\boldsymbol{F}_{\text{dsm}}^{\text{align}}$ are mapped into the frequency domain, decomposed into LF and HF components, and then undergo frequency compensation. Specifically, LF features are enhanced under DSM guidance to ensure elevation consistency, while HF features are refined under optical guidance to recover boundary and detail information. The compensated features are fused as $\boldsymbol{F}_{\text{fusion}}$ and projected back to the spatial domain. Finally, $\boldsymbol{F}_{\text{fusion}}$ is processed by a multiscale decoder $\boldsymbol{F}_{\text{Decoder}}$, which aggregates semantic cues at different resolutions and generates the final pixel-wise segmentation output $\boldsymbol{S}$.

**Algorithm 1** Overall Pipeline of GLFRNet

**Input:** Optical image $\boldsymbol{I}_{\text{opt}} \in \mathbb{R}^{3\times H\times W}$, DSM data $\boldsymbol{I}_{\text{dsm}} \in \mathbb{R}^{1\times H\times W}$

**Output:** Segmentation map $S$

```
//Feature Extraction
F_opt^e ← F_Encoder^opt(I_opt);
F_dsm^e ← F_Encoder^dsm(I_opt);
//Dual-Stage Encoding
F_opt^e ← DSAE(F_opt^e);
F_dsm^e ← DSAE(F_dsm^e);
//Global-Local Feature Alignment
F_opt^align, F_dsm^align ← FPAA(F_opt^e, F_dsm^e);
//Frequency Reconstruction
F_opt^freq, F_dsm^freq ← FFT(F_opt^align), FFT(F_dsm^align);
Split F_opt^freq, F_dsm^freq into low-frequency (LF) and high-frequency (HF) components;
LF ← LF compensation under DSM guidance;
HF ← HF compensation under Optical guidance;
F_fusion ← FreqFusion(LF, HF);
F_fusion ← IFFT(F_fusion);
//Multiscale Decoding
S ← F_Decoder(F_fusion);
return S;
```

This unified architecture enables GLFRNet to effectively preserve modality-specific advantages, achieve robust multimodal fusion, and deliver accurate semantic segmentation performance. The overall process is shown in Algorithm 1.

### *B. Dual-Stage Structure-Aware Encoder*

Within the proposed multimodal semantic segmentation framework, the DSAE serves as the backbone for hierarchical feature extraction, enabling effective synergy between optical and DSM modalities. As shown in Fig. 2, DSAE is designed to leverage the distinct characteristics of each modality while facilitating feature coordination across different semantic levels. Given registered optical and DSM images, $\boldsymbol{X}_{\text{opt}}$ and $\boldsymbol{X}_{\text{dsm}}$, we first standardize and independently feed them into two modality-specific ResNet [50] encoders, denoted as $\mathcal{F}_{\text{ResNet}}^{\text{opt}}$ and $\mathcal{F}_{\text{ResNet}}^{\text{dsm}}$. The optical branch focuses on capturing spectral texture and edge details, while the DSM branch emphasizes geometric and elevation-related cues. The initial stage outputs are given by

$$\boldsymbol{F}_{\text{opt}}^{(1)} = \mathcal{F}_{\text{ResNet}}^{\text{opt}}\left(\boldsymbol{X}_{\text{opt}}\right), \quad \boldsymbol{F}_{\text{dsm}}^{(1)} = \mathcal{F}_{\text{ResNet}}^{\text{dsm}}\left(\boldsymbol{X}_{\text{dsm}}\right) \tag{1}$$

where $\boldsymbol{F}_{\text{opt}}^{(1)}$ and $\boldsymbol{F}_{\text{dsm}}^{(1)}$ represent the first-stage semantic features, encompassing multiscale and spatially detailed information. Despite the effectiveness of ResNet encoders in extracting local details, their ability to model long-range semantic dependencies is limited, especially in complex scenes with structural ambiguity or cross-modal spatial misalignment. To enhance global context modeling, we introduce a second-stage encoder composed of GBC-Mamba blocks. This module aggregates global context and long-range dependencies by integrating several parallel attention pathways and bidirectional state-space modeling, enabling robust structural reasoning across modalities. For each modality, the first-stage features $\boldsymbol{F}_m^{(1)}$ ($m \in \{\text{opt}, \text{dsm}\}$) are first normalized and linearly projected

$$\boldsymbol{Z}_m = \text{Linear}\left(\text{Norm}\left(\boldsymbol{F}_m^{(1)}\right)\right). \tag{2}$$

The projected features $\boldsymbol{Z}_m$ are then refined by the GBC-Mamba block, which updates the hidden state as

$$H_t = A \cdot H_{t-1} + B \cdot Z_t + C \cdot H_{t-1} + D \cdot Z_t \tag{3}$$

where $H_t$ denotes the hidden state at time $t$, and $A, B, C, D$ are learnable parameters modeling temporal and contextual dependencies. The output of the second-stage encoder is as

$$\boldsymbol{F}_m^{(2)} = \mathcal{F}_{\text{GBC-Mamba}}\left(\boldsymbol{Z}_m\right). \tag{4}$$

To further mitigate spatial misalignment and improve geometric consistency, we introduce a cross-modal spatial attention mechanism. Specifically, the DSM features guide the optical features to enhance boundary delineation and suppress irrelevant textures. The attention gate is defined as

$$\boldsymbol{G}_x = \sigma\left(\text{Conv}\left([VX_{dsm}, \boldsymbol{F}_{\text{opt}}^{(1)}\right)\right) \tag{5}$$

where $VX_{dsm}$ denotes the spatial encoding of DSM features, $\sigma(\cdot)$ is the sigmoid activation, and $\text{Conv}(\cdot)$ denotes a $1 \times 1$ convolution. The gated output for the optical branch is

$$\boldsymbol{F}_{\text{opt}}^{(2)} = \boldsymbol{G}_x \odot \boldsymbol{F}_{\text{opt}}^{(2)} + (1 - \boldsymbol{G}_x) \odot \boldsymbol{F}_{\text{opt}}^{(1)}. \tag{6}$$

After dual-stage encoding, the outputs from both branches are concatenated and forwarded to the following alignment and reconstruction modules for further fusion and refinement:

$$\boldsymbol{F}_{\text{DSAE}} = \text{Concat}\left(\boldsymbol{F}_{\text{opt}}^{(2)}, \boldsymbol{F}_{\text{dsm}}^{(2)}\right). \tag{7}$$

The proposed DSAE not only preserves modality-specific advantages but also explicitly models multiscale spatial context and cross-modal structural dependencies.

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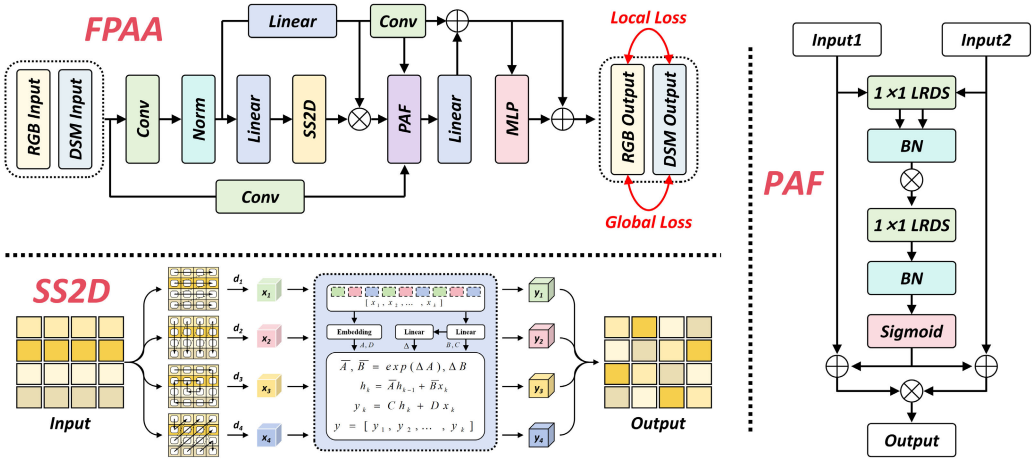



Fig. 3. Architecture of the proposed global–local FPAA. The FPAA integrates SS2D for global context modeling and PAF for local alignment, enabling effective cross-modal feature interaction and semantic consistency between optical and DSM modalities.

### C. Global–Local FPAA

In multimodal feature fusion, heterogeneous modalities such as optical and DSM data often exhibit significant statistical and structural discrepancies, leading to semantic misalignment and suboptimal fusion if not properly addressed. To mitigate this, we introduce the global–local FPAA, as shown in Fig. 3, which enables both global semantic consistency and local structure alignment across modalities via joint feature projection and adaptive interaction. Let $\boldsymbol{X}_{\text{opt}} \in \mathbb{R}^{C\times H\times W}$ and $\boldsymbol{X}_{\text{dsm}} \in \mathbb{R}^{C'\times H\times W}$ denote the input features from the optical and DSM branches, respectively. For a given modality $m \in \{\text{opt}, \text{dsm}\}$, the feature embedding is first normalized and projected as follows:

$$\boldsymbol{F}_m = \text{Norm}\left(\text{Conv}\left(X_m\right)\right). \tag{8}$$

The normalized features are then processed by a layer-wise linear transformation

$$\boldsymbol{F}_i = W_i \boldsymbol{F}_m + b_i \tag{9}$$

where $W_i$ and $b_i$ are learnable parameters. To enhance long-range dependency modeling, we incorporate a selective state space dynamics (SS2D) module, which efficiently captures nonlocal context through dynamic state updates in both spatial and temporal dimensions

$$h_t = Ah_{t-1} + Bx_t, \quad y_t = Ch_t + Dx_t \tag{10}$$

where $A$, $B$, $C$, and $D$ are system matrices, and $x_t$ is the projected input at time $t$. This state-space framework preserves sequential dependencies and nonlocal relationships, providing a robust basis for feature alignment. Next, a pixel-wise attention fusion (PAF) module [51] is introduced to achieve fine-grained local alignment by adaptively weighting features across modalities. Specifically, given the SS2D-encoded representations $H_1$ and $H_2$ from the two modalities, the attention map is calculated as

$$\boldsymbol{S} = \sigma\left(\phi\left(H_1, H_2\right)\right) \tag{11}$$

where $\phi(\cdot)$ denotes cross-modal interaction implemented via a lightweight LRDS block, and $\sigma(\cdot)$ is the sigmoid function. The fused feature is then given by

$$\boldsymbol{F}_{paf} = \boldsymbol{S} \odot \boldsymbol{Y}_a + (1 - \boldsymbol{S}) \odot \boldsymbol{Y}_b \tag{12}$$

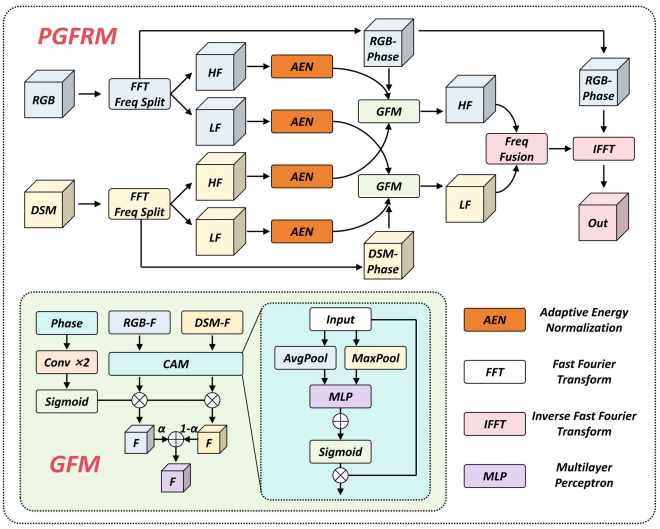



Fig. 4. Overview of the proposed PGFRM. The PGFRM performs modality-wise frequency decomposition, AEN, global frequency modulation, and phase-guided amplitude compensation, followed by inverse Fourier reconstruction to achieve effective cross-modal feature fusion.

where $Y_a$ and $Y_b$ are the transformed representations from each modality, and $\odot$ denotes element-wise multiplication. To further strengthen global–local consistency, the fused representations are aggregated through concatenation and convolutional projection, followed by a multilayer perceptron (MLP) for final feature alignment:

$$\boldsymbol{F}_{\text{FPAA}} = \text{MLP}\left(\left[\boldsymbol{Y}^m, \text{Conv}\left(\boldsymbol{F}_i\right), \text{Conv}\left(\boldsymbol{X}_m\right)\right]\right) \tag{13}$$

where $[\,\cdot\,]$ denotes concatenation across channels. The FPAA not only aligns semantic distributions but also adaptively compensates for structural discrepancies between modalities.

### D. Phase-Guided Frequency Reconstruction Module

In the process of multimodal feature fusion, optical and DSM modalities often provide complementary information in different frequency bands. The optical modality excels at conveying fine spectral and textural cues, whereas DSM emphasizes structural and elevation features, primarily distributed in the LF domain. However, direct concatenation

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or fusion in the spatial domain may lead to information redundancy and suboptimal integration. To address this, we propose the PGFRM, as shown in Fig. 4, which performs cross-modal frequency decomposition, adaptive normalization, and phase-guided compensation.

Given the fused features from the previous stage, $\boldsymbol{F}_{\text{opt}} \in \mathbb{R}^{C\times H\times W}$ and $\boldsymbol{F}_{\text{dsm}} \in \mathbb{R}^{C'\times H\times W}$, we first transform them into the frequency domain using FFT

$$\boldsymbol{F}_{\text{opt}}^{\text{freq}} = \text{FFT}\left(\boldsymbol{F}_{\text{opt}}\right), \quad \boldsymbol{F}_{\text{dsm}}^{\text{freq}} = \text{FFT}\left(\boldsymbol{F}_{\text{dsm}}\right). \tag{14}$$

The frequency representations are decomposed into amplitude and phase components as follows:

$$\boldsymbol{A}_{\text{opt}}, \boldsymbol{P}_{\text{opt}}, \boldsymbol{A}_{\text{dsm}}, \boldsymbol{P}_{\text{dsm}} = \text{AmpPhase}\left(\boldsymbol{F}_{\text{opt}}^{\text{freq}}, \boldsymbol{F}_{\text{dsm}}^{\text{freq}}\right) \tag{15}$$

where $A$ and $P$ denote the amplitude and phase spectra, respectively. We further split the amplitude spectrum into HF and LF components

$$\boldsymbol{A}_{\text{opt}}^{\text{HF}}, \boldsymbol{A}_{\text{opt}}^{\text{LF}}, \boldsymbol{A}_{\text{dsm}}^{\text{HF}}, \boldsymbol{A}_{\text{dsm}}^{\text{LF}} = \text{Split}\left(\boldsymbol{A}_{\text{opt}}, \boldsymbol{A}_{\text{dsm}}\right). \tag{16}$$

The HF components preserve fine-grained details, while the LF components capture global structure. Adaptive energy normalization (AEN) is applied to each frequency band to mitigate scale discrepancies and enhance robustness

$$\tilde{A} = \frac{A}{\sqrt{\alpha T + \beta E + \epsilon}} \tag{17}$$

where $T$ and $E$ denote the total and average energy distributions for each frequency band, $\alpha$, $\beta$, and $\epsilon$ are learnable parameters. To achieve dynamic global frequency modulation, we aggregate the normalized amplitudes through both average and max pooling, followed by an MLP:

$$v_e = \text{MLP}\left(\text{AvgPool}\left(A\right) + \text{MaxPool}\left(A\right)\right), \quad \hat{A} = v_e \odot A \tag{18}$$

where $v_e$ represents a learned global modulation vector. For adaptive frequency compensation, we introduce a phase-guided mechanism (PG). The difference between the phase spectra of optical and DSM features is calculated as

$$\Delta P = \tanh\left(\text{Conv}\left(\boldsymbol{P}_{\text{opt}} - \boldsymbol{P}_{\text{dsm}}\right)\right). \tag{19}$$

The phase-guided compensation of the amplitude spectra is then formulated as

$$A_{comp} = (1 - \Delta P) \odot \hat{A}_{opt} + \Delta P \odot \hat{A}_{dsm}. \tag{20}$$

Finally, the compensated amplitude and the original phase are fused, and the inverse FFT (IFFT) is applied to reconstruct the spatial-domain features

$$\boldsymbol{F}_{\text{PGFRM}} = \text{IFFT}\left(A_{comp}, P_{opt}\right). \tag{21}$$

The PGFRM adaptively decomposes, normalizes, and reconstructs multimodal features in the frequency domain under phase guidance, which achieves cross-modal information integration and suppresses redundant responses.

### E. Loss Function

During network training, we employ a composite optimization objective that unifies the primary semantic segmentation loss with an auxiliary cross-modal alignment loss, thereby ensuring both frequency-domain precision and spatial-structural consistency between optical and DSM modalities

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ce}}\mathcal{L}_{\text{ce}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}} \tag{22}$$

where $\mathcal{L}_{\text{ce}}$ is the semantic segmentation loss, $\mathcal{L}_{\text{align}}$ is the cross-modal alignment loss (introduced by FPAA), and $\lambda_{\text{ce}} = 1.0$, $\lambda_{\text{align}} = 0.5$ are the weighting coefficients. We use a class-weighted cross-entropy loss to supervise the output

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{C} y_{i,k} \log p_{i,k} \tag{23}$$

where $N$ is the number of pixels, $C$ is the number of classes, $y_{i,k}$ denotes ground-truth label, and $p_{i,k}$ is the predicted class probability for pixel $i$. To encourage spatial and frequency consistency between optical and DSM features, we introduce a hybrid alignment loss as follows:

$$\mathcal{L}_{\text{align}} = \lambda_g\mathcal{L}_{\text{global}} + \lambda_l\mathcal{L}_{\text{local}} \tag{24}$$

where $\lambda_g = 0.5$ and $\lambda_l = 0.5$ are the hyperparameters. The global alignment loss measures the overall spatial similarity using global average pooling (GAP) as follows:

$$\mathcal{L}_{\text{global}} = \left\|\text{GAP}\left(\boldsymbol{F}_{\text{opt}}^{\text{out}}\right) - \text{GAP}\left(\boldsymbol{F}_{\text{dsm}}^{\text{out}}\right)\right\|_1 \tag{25}$$

where $\boldsymbol{F}_{\text{opt}}^{\text{out}}$ and $\boldsymbol{F}_{\text{dsm}}^{\text{out}}$ are the fused features of each modality after FPAA. The local alignment loss enforces patch-wise correspondence to preserve local spatial consistency as

$$\mathcal{L}_{\text{local}} = \frac{1}{M}\sum_{j=1}^{M} w_j^{\text{edge}} \left\| f_j^{\text{opt}} - f_j^{\text{dsm}} \right\|_1 \tag{26}$$

where $f_j^{\text{opt}}$ and $f_j^{\text{dsm}}$ are feature vectors for the $j$th patch, $M$ is the number of sampled patches, and $w_j^{\text{edge}}$ is an edge-aware weighting factor to emphasize boundaries and fine structures.

## IV. Experiments and Analysis

### A. Dataset Description

*1) ISPRS Potsdam Dataset [52]:* This dataset provides high-resolution (5 cm) aerial images with pixel-level annotations for six semantic classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. Each image contains four multispectral bands (infrared, red, green, and blue) and a normalized DSM. In our experiments, we utilize only the RGB bands and DSM. The dataset consists of 38 images (6000 × 6000 pixels), of which 14 are used for training and 24 for testing. Images are randomly cropped into 512 × 512 patches, yielding 140 training samples.

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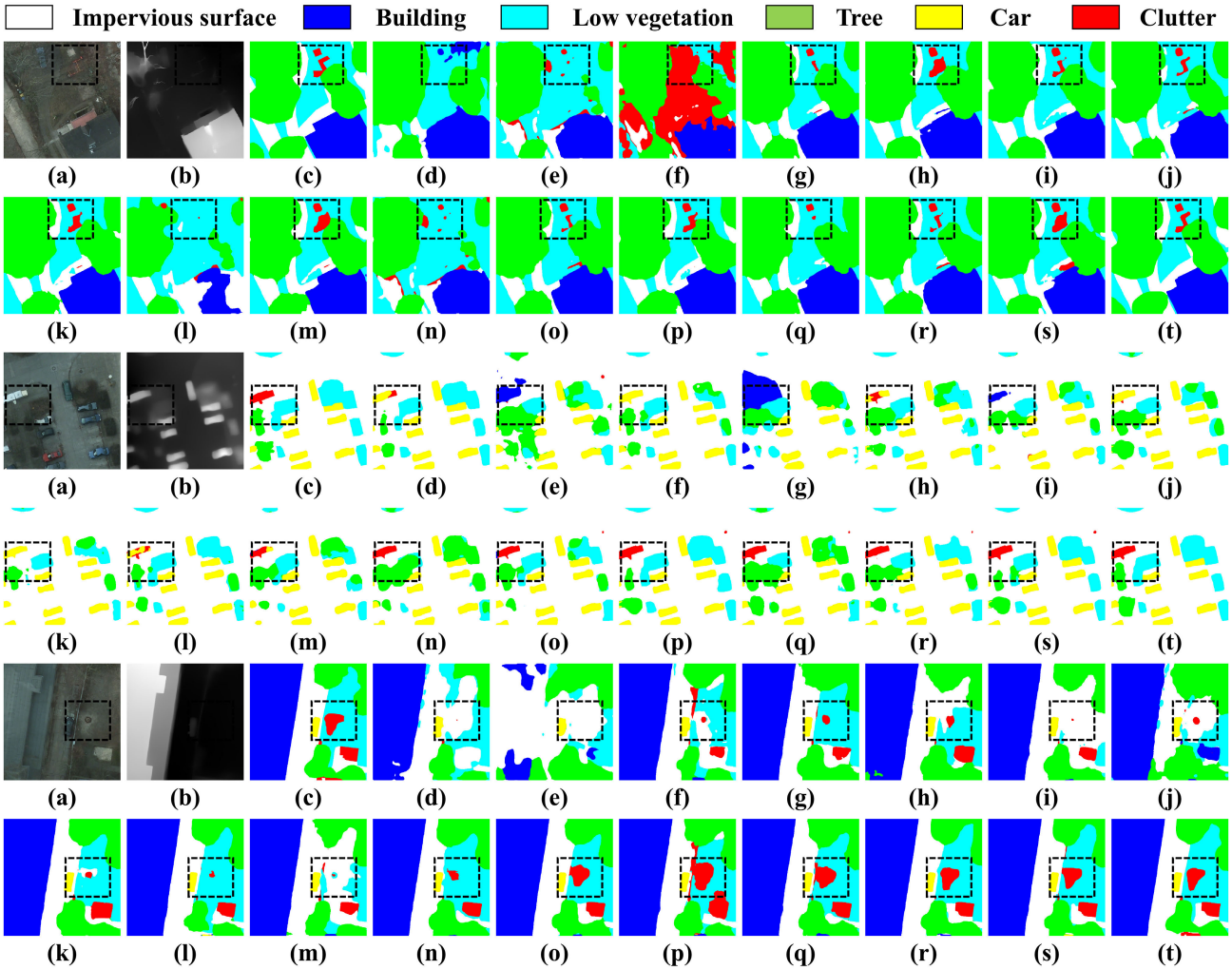



Fig. 5. Qualitative comparison of the visualization results of our method and other approaches on the ISPRS Potsdam dataset. (a) Optical. (b) DSM. (c) GT. (d) DeepLabV3+. (e) PSPNet. (f) UNetFormer. (g) RS3Mamba. (h) TransUNet. (i) CMFNet. (j) FuseNet. (k) ESANet. (l) SA-GATE. (m) CMGFNet. (n) MIEFNet. (o) MultiSenseSeg. (p) MFTransNet. (q) FTransUNet. (r) FTransDeepLab. (s) MFNet. (t) GLFRNet (Ours).

*2) ISPRS Vaihingen Dataset [53]:* This dataset contains 33 high-resolution (9 cm) aerial images (2500 × 2500 pixels), annotated for the same six semantic categories. Each image includes three spectral bands (near-infrared, red, green) and a normalized DSM. The dataset is split into 17 training and 16 test images. All images are partitioned into 512 × 512 patches to facilitate model training and evaluation. Both datasets are preprocessed to ensure compatibility with deep multimodal architectures, and only the RGB (or NIRRG) bands together with DSM are used in all experiments.

*3) VALID Dataset [54]:* The VALID dataset provides a large-scale and challenging benchmark for multimodal remote sensing semantic segmentation, featuring 1350 pairs of high-resolution optical and DSM images (1024 × 1024 pixels). For efficient training, images are cropped into 256 × 256 patches, yielding 21 600 pairs. The dataset covers over 30 categories, including both common (building, tree, and road) and fine-grained classes (bridge, land, and low obstacle), and is characterized by significant interclass ambiguity and incomplete modality coverage, reflecting real-world remote sensing complexity. The data are split into 70% for training (15 120 pairs), 10% for validation (2160), and 20% for testing (4320).

## B. Implementation Details

All experiments are conducted on a workstation with an Intel Xeon E5-2683 v4 CPU, NVIDIA RTX A5000 GPU (24 GB VRAM), and Ubuntu 20.04. The software stack includes Python 3.10, PyTorch 1.13.1, and CUDA 12.3. To improve generalization under limited data, we apply data augmentation techniques such as random cropping, flipping, and color jittering. Training employs the Adam optimizer with an initial learning rate of $1 \times 10^{-4}$, weight decay of $5 \times 10^{-4}$, and a cosine annealing schedule [55] over 300 epochs. All images are resized to 512 × 512 pixels, with a batch size of 8. During inference, models predict at the same resolution as training. The output results are upsampled to the original image size for pixel-level evaluation and visualization.

## C. Evaluation Metrics

To comprehensively assess the performance of semantic segmentation models, three widely adopted evaluation metrics are employed: overall accuracy (OA), mean $F1$-score (m$F1$), and mean intersection-over-union (mIoU).

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TABLE I

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS POTSDAM DATASET. THE OPTIMAL VALUES IN EACH COLUMN ARE IN BOLD AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| Type | Method | OA(%) | | | | | | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|---|---|---|---|
| | | Bui. | Tre. | Low. | Car | Imp. | Total | | |
| Single Modal | DeepLabV3+ | 97.16 | 84.20 | 83.73 | 89.49 | 90.55 | 89.02 | 88.29 | 81.18 |
| | PSPNet | 97.03 | 83.13 | 85.67 | 88.81 | 90.91 | 88.67 | 88.92 | 80.36 |
| | UNetFormer | 97.69 | 86.47 | 87.93 | 95.91 | 92.27 | 90.65 | 91.71 | 85.05 |
| | RS$^3$Mamba | 97.70 | 86.11 | 89.53 | 96.23 | 91.36 | 90.49 | 91.69 | 85.01 |
| Multimodal | TransUNet | 96.63 | 82.65 | 82.98 | 93.17 | 91.93 | 90.01 | 90.97 | 83.74 |
| | CMFNet | 97.63 | 87.40 | 88.00 | 95.68 | 92.84 | 91.16 | 92.10 | 85.63 |
| | FuseNet | 97.48 | 85.14 | 85.71 | 96.10 | 92.64 | 90.58 | 91.60 | 84.86 |
| | ESANet | 97.10 | 85.31 | 87.41 | 94.08 | 92.76 | 89.74 | 91.22 | 84.15 |
| | SA-GATE | 96.54 | 81.18 | 85.31 | 96.63 | 90.77 | 87.91 | 90.26 | 82.53 |
| | CMGFNet | 97.41 | 86.80 | 86.80 | 95.68 | 92.60 | 90.21 | 91.40 | 84.53 |
| | MIEFNet | 97.65 | 86.77 | 87.93 | 95.30 | 91.35 | 90.50 | 91.46 | 84.61 |
| | MultiSenseSeg | 97.94 | 87.27 | 84.22 | 95.24 | **93.88** | 91.16 | 92.01 | 85.53 |
| | MFTransNet | 97.37 | 85.81 | 86.92 | 96.05 | 92.45 | 89.96 | 91.11 | 84.04 |
| | FTransUNet | 97.78 | 88.22 | 87.36 | 96.31 | 93.17 | 91.34 | 92.41 | 86.20 |
| | FTransDeepLab | 97.58 | 85.87 | 89.07 | 96.94 | 92.81 | 90.97 | 92.08 | 85.62 |
| | MFNet | 98.44 | 87.37 | **90.36** | 96.24 | 93.14 | 91.71 | 92.70 | 86.69 |
| | GLFRNet | **98.62** | **88.41** | 90.15 | **96.97** | 93.86 | **91.93** | **92.83** | **87.03** |

1) OA measures the proportion of correctly classified pixels across the entire test set

$$\mathrm{OA} = \frac{TP + TN}{TP + TN + FP + FN} \tag{27}$$

where $TP$, $TN$, $FP$, and $FN$ denote the number of true positives, true negatives, false positives, and false negatives.

2) $F1$-score is the harmonic mean of precision and recall for each class as follows:

$$\mathrm{F1}_i = \frac{2 \cdot \mathrm{Precision}_i \cdot \mathrm{Recall}_i}{\mathrm{Precision}_i + \mathrm{Recall}_i} \tag{28}$$

$$\mathrm{Precision}_i = \frac{TP_i}{TP_i + FP_i}, \quad \mathrm{Recall}_i = \frac{TP_i}{TP_i + FN_i}. \tag{29}$$

The m$F1$ is then calculated as

$$\mathrm{mF1} = \frac{1}{C}\sum_{i=1}^{C} \mathrm{F1}_i \tag{30}$$

where $C$ is the total number of classes.

3) mIoU evaluates the spatial overlap between predicted and ground truth segments

$$\mathrm{IoU}_i = \frac{|P_i \cap G_i|}{|P_i \cup G_i|}, \quad \mathrm{mIoU} = \frac{1}{C}\sum_{i=1}^{C} \mathrm{IoU}_i \tag{31}$$

where $P_i$ and $G_i$ denote the predicted and ground truth pixel sets for class $i$, and $|\cdot|$ denotes set cardinality.

### D. Comparison With State-of-the-Art Methods

To demonstrate the effectiveness of the proposed GLFRNet, we conduct comprehensive evaluations on three benchmark test sets. The comparison includes both single-modality methods, such as DeepLabV3+ [18], PSPNet [19], UNetFormer [24], and RS3Mamba [37], as well as recent multimodality approaches, including TransUNet [29], CMFNet [30], FuseNet [12], ESANet [31], SA-GATE [32], CMGFNet [13], MIEFNet [40], MultiSenseSeg [41], MFTransNet [42], FTransUNet [43], FTransDeepLab [44], and MFNet [45]. For fair comparison, all models are evaluated on test images, and consistent with the training configuration.

*1) Results on ISPRS Potsdam Dataset:* Table I presents the quantitative comparison between the proposed GLFRNet and a range of state-of-the-art methods on the ISPRS Potsdam test set. GLFRNet achieves the highest performance across all major metrics, with an OA of 98.62%, m$F1$ of 92.83%, and mIoU of 87.03%. Compared with the strongest competing multimodal baseline, MFNet, GLFRNet improves mIoU by 0.34% and m$F1$ by 0.13%, demonstrating superior capability in global feature modeling and cross-modal information fusion. In terms of per-class segmentation, GLFRNet consistently outperforms other methods, especially for challenging categories such as building, low vegetation, and car. Specifically, the m$F1$ for the building class reaches 90.15%, exhibiting strong structural consistency even in complex shadow regions and along object boundaries. For the car class, GLFRNet attains an m$F1$ of 96.97%, outperforming MFNet by 0.73%, which highlights the effective exploitation of DSM-derived height cues to enhance the recognition of small and occluded objects. Fig. 5 provides qualitative comparisons of segmentation results. Traditional single-modality methods such as DeepLabV3+ and PSPNet often suffer from blurred object boundaries and misclassifications in shadowed or highly reflective areas. While existing multimodal methods exhibit improvements in structural delineation, some (e.g., FuseNet and ESANet) still present noise artifacts and class confusion, particularly at building edges and road intersections. In contrast, GLFRNet produces segmentation maps with noticeably smoother and more continuous contours, better edge preservation, and higher boundary consistency, especially in scenes with significant elevation variation (e.g., cars and buildings). Furthermore, GLFRNet demonstrates superior discriminative capability in complex scenarios involving mixed land covers, such as overlapping grass and low vegetation,

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TABLE II

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS VAIHINGEN DATASET. THE OPTIMAL VALUES IN EACH COLUMN ARE IN BOLD AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| Type | Method | OA(%) | | | | | | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|---|---|---|---|
| | | Bui. | Tre. | Low. | Car | Imp. | Total | | |
| Single Modal | DeepLabV3+ | 93.74 | 91.41 | 78.11 | 80.14 | 91.40 | 86.96 | 87.36 | 78.03 |
| | PSPNet | 94.52 | 90.17 | 78.84 | 79.22 | 92.03 | 89.94 | 86.55 | 76.96 |
| | UNetFormer | 96.23 | 91.85 | 79.95 | 86.99 | 91.85 | 91.17 | 89.85 | 81.97 |
| | RS$^3$Mamba | 97.40 | 92.14 | 79.56 | 88.15 | 92.19 | 91.64 | 90.34 | 82.78 |
| Multimodal | TransUNet | 96.48 | 92.77 | 76.14 | 69.56 | 91.66 | 90.96 | 87.34 | 78.26 |
| | CMFNet | 97.17 | 90.82 | 80.37 | 85.47 | 92.36 | 91.40 | 89.48 | 81.44 |
| | FuseNet | 96.28 | 90.28 | 78.98 | 81.37 | 91.66 | 90.51 | 87.71 | 78.71 |
| | ESANet | 95.69 | 90.50 | 77.16 | 85.46 | 91.39 | 90.61 | 88.18 | 79.42 |
| | SA-GATE | 94.84 | 92.56 | 81.29 | 87.79 | 91.69 | 91.10 | 89.81 | 81.27 |
| | CMGFNet | 97.75 | 91.60 | 80.03 | 87.28 | 92.35 | 91.72 | 90.00 | 82.26 |
| | MIEFNet | 98.35 | 93.17 | 69.16 | 88.30 | 90.69 | 90.30 | 87.51 | 78.42 |
| | MultiSenseSeg | 96.79 | 93.20 | 77.98 | 90.99 | 91.82 | 91.47 | 89.94 | 82.15 |
| | MFTransNet | 96.41 | 91.48 | 80.09 | 86.52 | 92.11 | 91.22 | 89.62 | 81.61 |
| | FTransUNet | 98.20 | 91.94 | **81.49** | 91.27 | 92.40 | 91.21 | 91.21 | 84.23 |
| | FTransDeepLab | 98.11 | 93.45 | 80.35 | 89.98 | 93.23 | 92.61 | 91.00 | 83.87 |
| | MFNet | 98.38 | **93.94** | 80.70 | 90.47 | 93.59 | 92.97 | 91.71 | 85.03 |
| | GLFRNet | **98.63** | 93.68 | 80.68 | **91.88** | **93.62** | **93.27** | **92.33** | **85.35** |

indicating robust generalization to diverse surface structures. The superior performance of GLFRNet stems from the integration of dual-encoder and Mamba, which together enhance cross-modal consistency and global feature modeling. These innovations enable GLFRNet to achieve state-of-the-art results in both quantitative and qualitative evaluations on remote sensing segmentation.

*2) Results on ISPRS Vaihingen Dataset:* Table II summarizes the quantitative results of GLFRNet and various state-of-the-art single-modal and multimodal methods on the ISPRS Vaihingen test set. GLFRNet consistently achieves the best performance across all primary metrics, with an OA of 98.63%, m$F1$ of 92.33%, and mIoU of 85.35%. Compared with the strongest baseline MFNet, GLFRNet provides improvements of 0.25% in OA, 0.62% in m$F1$, and 0.32% in mIoU, further validating its superiority in multimodal feature representation and semantic consistency modeling. Examining the per-class results, GLFRNet excels in categories such as building, car, and impervious surface. Notably, for the car class, GLFRNet achieves an m$F1$ of 91.88%, outperforming FTransUNet (91.27%) and MFNet (90.47%), which demonstrates enhanced robustness in small object detection and segmentation of highly reflective regions. This improvement can be attributed to the PG, which enables effective alignment of energy distributions and structural features across modalities, thereby reducing semantic drift caused by texture conflicts or geometric misalignments. Furthermore, GLFRNet maintains stable performance in the low vegetation and tree classes (80.68% and 93.68% m$F1$, respectively, even under challenging illumination and shadow conditions. Compared to advanced single-modal architectures such as RS$^3$Mamba and UNetFormer, GLFRNet improves mIoU by more than 3%, indicating that the FPAA effectively mitigates cross-modal distribution inconsistencies and strengthens the synergy between high-level semantic and spatial features.

TABLE III

QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE VALID DATASET. THE OPTIMAL VALUES IN EACH COLUMN ARE IN BOLD AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| Type | Method | OA(%) | mF1(%) | mIoU(%) |
|---|---|---|---|---|
| Single Modal | DeepLabV3+ | 85.43 | 82.17 | 72.36 |
| | PSPNet | 86.45 | 84.29 | 74.18 |
| | UNetFormer | 87.59 | 86.41 | 75.68 |
| | RS$^3$Mamba | 88.17 | 87.63 | 76.53 |
| Multimodal | TransUNet | 88.41 | 87.14 | 77.69 |
| | CMFNet | 89.09 | 87.97 | 78.21 |
| | FuseNet | 88.66 | 87.55 | 77.85 |
| | ESANet | 88.03 | 87.48 | 77.03 |
| | SA-GATE | 88.85 | 87.81 | 78.05 |
| | CMGFNet | 89.17 | 88.20 | 78.31 |
| | MIEFNet | 89.02 | 88.01 | 78.09 |
| | MultiSenseSeg | 89.18 | 88.34 | 78.47 |
| | MFTransNet | 90.28 | 90.21 | 82.01 |
| | FTransUNet | 90.58 | 91.02 | 84.36 |
| | FTransDeepLab | 90.13 | 90.58 | 83.23 |
| | MFNet | 90.57 | 90.82 | 84.10 |
| | GLFRNet | **91.22** | **92.01** | **85.21** |

Fig. 6 provides qualitative comparisons of segmentation outputs. Conventional single-modal methods (e.g., DeepLabV3 + and PSPNet) tend to exhibit blurred boundaries and misclassifications in regions with complex object edges or significant elevation changes, such as building shadows, road intersections, and occluded trees. Although multimodal methods improve structural delineation, some approaches (e.g., FuseNet and ESANet) still suffer from semantic confusion in high-reflectance or low-contrast areas. In contrast, GLFRNet produces segmentation maps with superior structural integrity and boundary continuity. The local zoom-in regions highlighted by black dashed boxes further illustrate the advantages in capturing building outlines, vehicle boundaries, and transitions between vegetation types.

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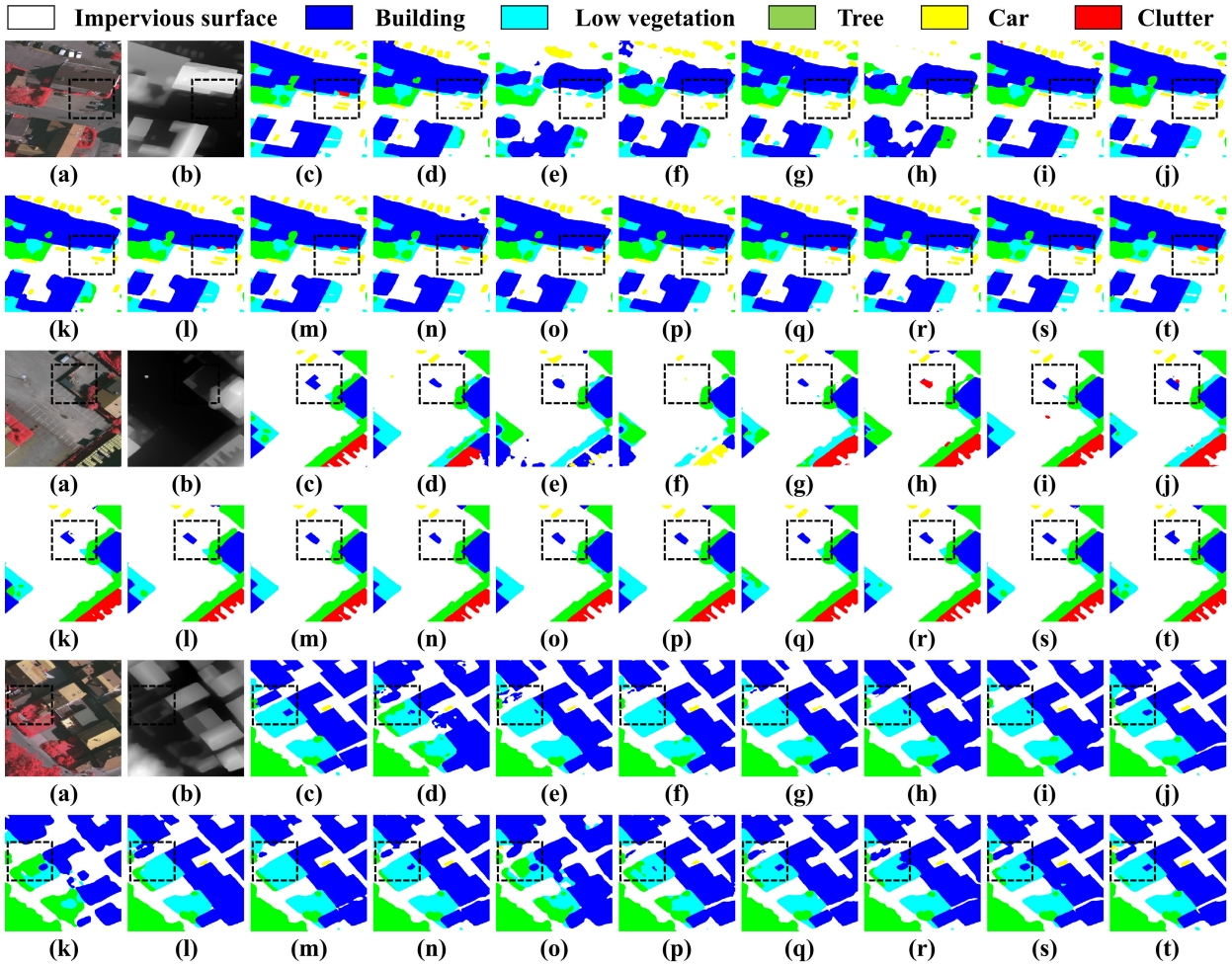



Fig. 6. Qualitative comparison of the visualization results of our method and other approaches on the ISPRS Vaihingen dataset. (a) Optical. (b) DSM. (c) GT. (d) DeepLabV3+. (e) PSPNet. (f) UNetFormer. (g) RS3Mamba. (h) TransUNet. (i) CMFNet. (j) FuseNet. (k) ESANet. (l) SA-GATE. (m) CMGFNet. (n) MIEFNet. (o) MultiSenseSeg. (p) MFTransNet. (q) FTransUNet. (r) FTransDeepLab. (s) MFNet. (t) GLFRNet (Ours).

Its predictions accurately recover geometric details of rooftops and provide precise localization of small objects with minimal omissions or class merging. These results are largely attributed to the PGFRM, which reconstructs energy distributions in the frequency domain, enabling efficient fusion of complementary information from VIS and DSM data under phase constraints. This not only enhances structural sensitivity and detail preservation but also improves semantic discrimination and contextual consistency in mixed land-cover areas.

*3) Results on VALID Dataset:* To thoroughly validate the generalization and robustness of our approach in highly diverse and realistic remote sensing scenarios, we further evaluate our model on the VALID dataset, which covers a wide range of urban and suburban landscapes, over 30 semantic categories, and heterogeneous data sources with different sensor modalities and spatial resolutions. As shown in Table III, GLFRNet achieves the best results among all compared methods, obtaining an mIoU of 85.21%. This represents a substantial improvement over the best single-modal baseline, RS$^3$Mamba (76.53%), as well as over the strongest multimodal competitors, including FTransUNet (84.36%), MFNet (84.10%), and MFTransNet (82.01%), highlighting the proposed method's strong capacity to handle complex category systems and modality variations. Notably, our approach achieves these gains not only in overall performance but also exhibits clear benefits in categories characterized by ambiguous appearance, fine spatial granularity, or severe occlusion, such as roads, bridges, and small-scale urban features, where many competing methods tend to generate fragmented or inconsistent predictions. Qualitative results shown in Fig. 7 further support these findings. GLFRNet demonstrates superior boundary-preserving capability and better semantic consistency across both common and rare classes, even in challenging scenes with mixed object distributions and significant intraclass variability. The superior performance on VALID, which is much more diverse and challenging than Potsdam and Vaihingen, convincingly demonstrates that our model not only fits classical urban aerial domains but also generalizes effectively to multi-source and fine-grained remote sensing analyses. These results underscore the practical potential of GLFRNet for robust and accurate semantic segmentation in real-world environments.

## E. Complexity Comparison

To assess the efficiency and deployability of GLFRNet, a comprehensive complexity analysis is conducted on the ISPRS Vaihingen dataset, benchmarking against representative

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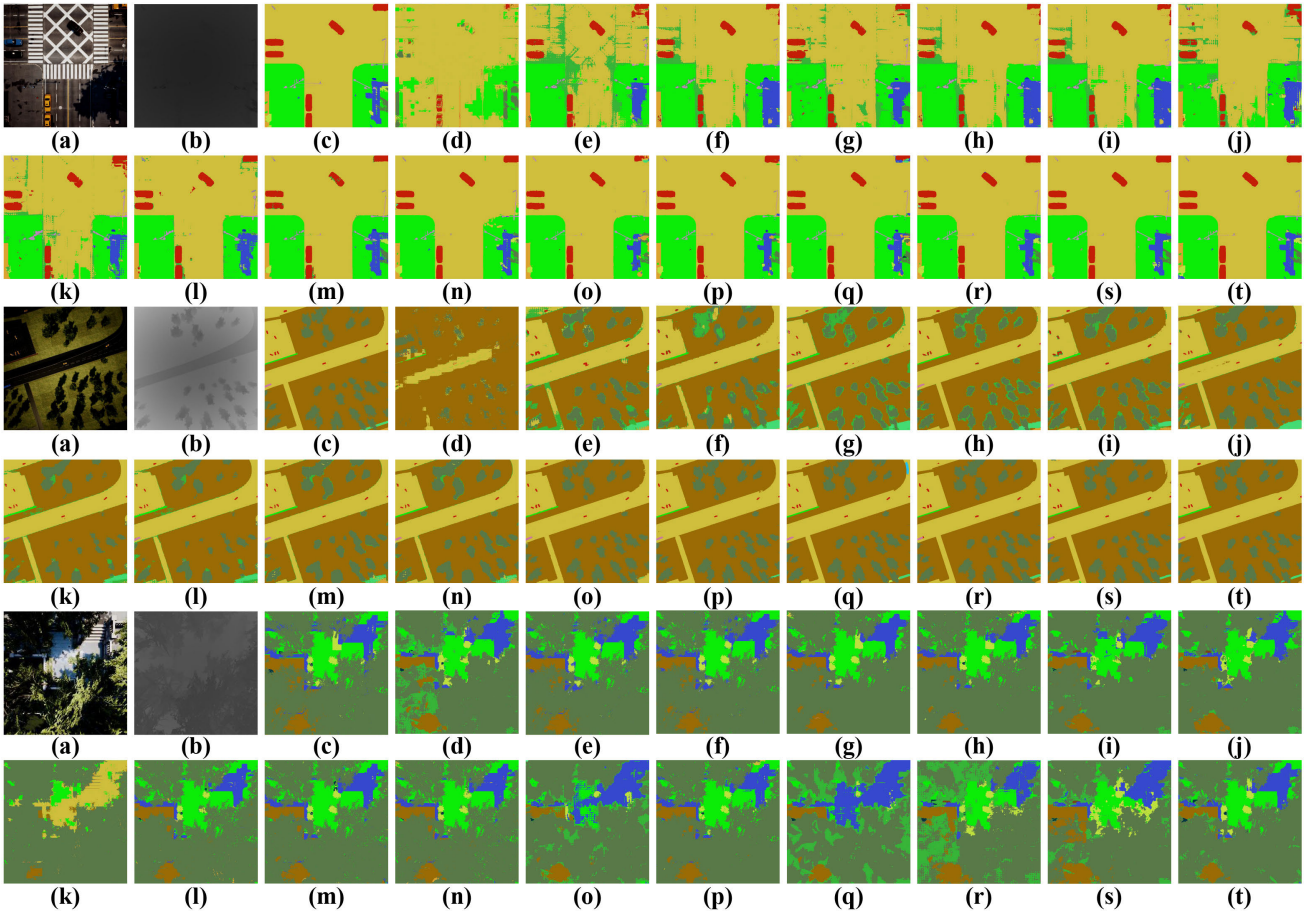



Fig. 7. Qualitative comparison of the visualization results of our method and other approaches on the VALID dataset. (a) Optical. (b) DSM. (c) GT. (d) DeepLabV3+. (e) PSPNet. (f) UNetFormer. (g) RS3Mamba. (h) TransUNet. (i) CMFNet. (j) FuseNet. (k) ESANet. (l) SA-GATE. (m) CMGFNet. (n) MIEFNet. (o) MultiSenseSeg. (p) MFTransNet. (q) FTransUNet. (r) FTransDeepLab. (s) MFNet. (t) GLFRNet (Ours).

TABLE IV

COMPARISON OF COMPUTATIONAL COMPLEXITY FOR DIFFERENT METHODS ON THE ISPRS VAIHINGEN DATASET. ALL METRICS ARE MEASURED ON NVIDIA RTX A5000 GPU WITH BATCH SIZE 8 AND 512 × 512 INPUT

| Type | Method | FLOPs (G)↓ | Params (M)↓ | Memory (MB)↓ | FPS↑ | mIoU (%) |
|---|---|---|---|---|---|---|
| Single Modal | DeepLabV3+ | 33.62 | 39.14 | 1650 | 78.27 | 78.03 |
| | PSPNet | 49.03 | 46.72 | 3124 | 66.01 | 76.96 |
| | UNetFormer | 6.04 | 24.20 | 1980 | 13.89 | 81.97 |
| | RS$^3$Mamba | 31.65 | 43.32 | 2332 | 26.12 | 82.78 |
| Multimodal | TransUNet | 32.27 | 93.23 | 3028 | 10.81 | 78.26 |
| | CMFNet | 78.25 | 123.63 | 4058 | 8.62 | 81.44 |
| | FuseNet | 58.37 | 42.08 | 2284 | 18.92 | 78.71 |
| | ESANet | 7.73 | 34.03 | 1914 | 10.42 | 79.42 |
| | SA-GATE | 41.28 | 110.85 | 3174 | 10.00 | 81.27 |
| | CMGFNet | 19.51 | 64.20 | 2463 | 11.61 | 82.26 |
| | MIEFNet | 35.33 | 127.54 | 3276 | 7.22 | 78.42 |
| | MultiSenseSeg | 67.21 | 186.77 | 3921 | 4.81 | 82.15 |
| | MFTransNet | 8.44 | 43.77 | 1549 | 14.88 | 81.61 |
| | FTransUNet | 45.21 | 160.88 | 3463 | 9.74 | 84.23 |
| | FTransDeepLab | 12.94 | 69.86 | 1624 | 15.02 | 83.87 |
| | MFNet | 42.85 | 111.28 | 6854 | 9.61 | 85.03 |
| | GLFRNet | 36.50 | 118.47 | 3961 | 13.75 | 85.35 |

single-modal and multimodal segmentation methods. All experiments are performed with unified hardware (NVIDIA RTX A5000 GPU, batch size 8, 512×512 input) to ensure fair comparison. Table IV presents key metrics, including FLOPs, parameters, memory usage, FPS, and mIoU. Single-modal methods such as DeepLabV3+, PSPNet, and UNetFormer

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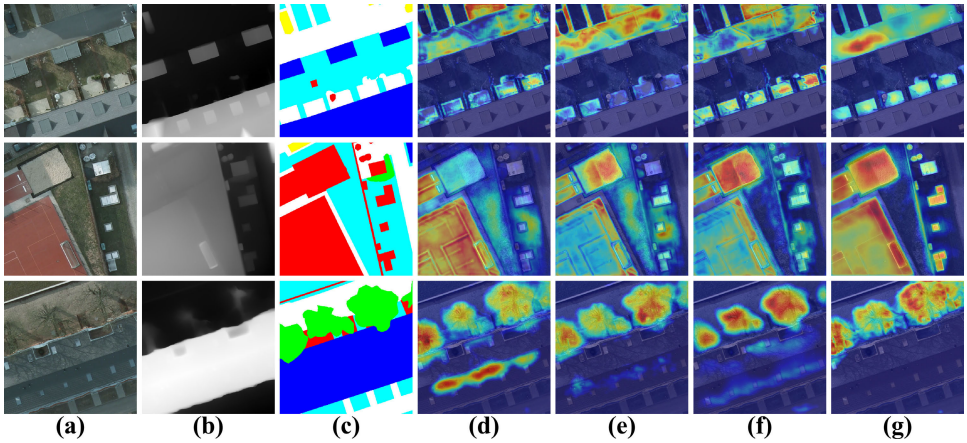



Fig. 8. Comparison of segmentation results and attention maps for different model structures on the ISPRS Potsdam test set. (a) Optical. (b) DSM. (c) GT. (d) Baseline + DSAE. (e) Baseline + FPAA. (f) Baseline + PGFRM. (g) GLFRNet.

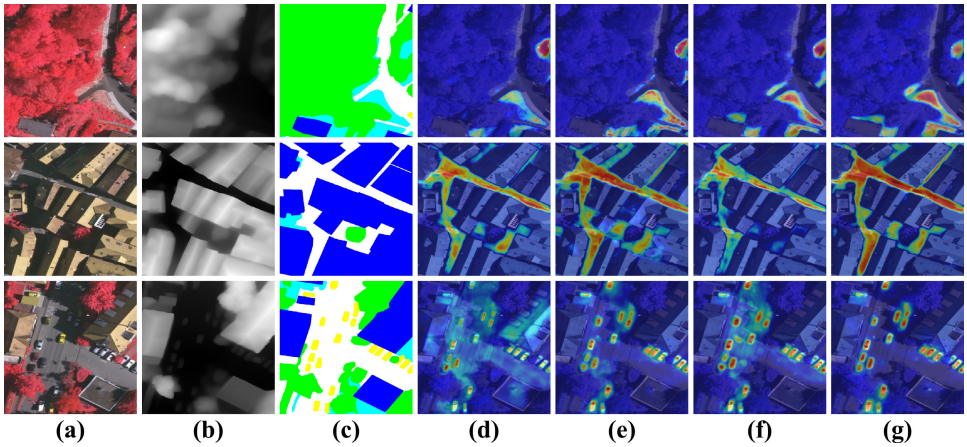



Fig. 9. Comparison of segmentation results and attention maps for different module combinations on the ISPRS Vaihingen test set. (a) Optical. (b) DSM. (c) GT. (d) w/o DSAE. (e) w/o FPAA. (f) w/o PGFRM. (g) GLFRNet (Ours).

exhibit lower computational demands but show limited segmentation performance in complex urban scenes due to the lack of multimodal input. Multimodal approaches, by incorporating both optical and DSM features, achieve substantial performance gains at the cost of increased resource consumption. Methods such as CMFNet and SA-GATE exceed 100-M parameters and 3 GB of memory, with FPS values consistently under 11. MFNet, representing a strong multimodal baseline, achieves an mIoU of 85.03% but with higher computational costs (111.28-M parameters, 6854-MB memory, 9.61 FPS). GLFRNet surpasses MFNet in accuracy (85.35% mIoU), while maintaining moderate resource requirements (118.47-M parameters, 3961-MB memory, 36.50-G FLOPs) and higher inference speed (13.75 FPS). These results demonstrate that GLFRNet achieves state-of-the-art segmentation performance with a balanced accuracy-efficiency tradeoff, supporting practical application in large-scale or resource-constrained scenarios.

### *F. Ablation Studies*

To rigorously assess the effectiveness of each component in GLFRNet, we conduct ablation studies on both the ISPRS Potsdam and ISPRS Vaihingen datasets. The analysis covers combinations of key structural modules, feature extraction blocks, and fusion strategies. All settings are kept consistent to ensure fairness and reproducibility.

*1) Effect of Structural Modules:* Table V summarizes the quantitative ablation results on ISPRS Potsdam. The baseline model with simple additive fusion achieves an mIoU of 81.33%. Introducing DSAE improves spatial alignment across modalities, raising mIoU to 84.76%. Adding FPAA further enhances global feature integration via frequency-domain attention, yielding an mIoU of 85.31%. Incorporating PGFRM boosts high-level semantic consistency, with mIoU reaching 84.52%. When all modules are combined, GLFRNet achieves the best performance, with OA 91.93%, m$F1$ 92.83%, and mIoU 87.03%, demonstrating strong complementarity and synergy among the modules. Fig. 8 shows qualitative results. The baseline exhibits blurred boundaries and missing details, especially at building and vegetation edges. DSAE increases class separability, but some local errors persist. FPAA sharpens edge definition and regional coherence, while PGFRM improves semantic completeness. The full GLFRNet restores details and continuity most accurately, closely matching the ground truth. Similar trends are observed on ISPRS Vaihingen (Table VI), where we further report the relative computational cost alongside accuracy. Removing DSAE, FPAA, or PGFRM reduces mIoU to 83.33%, 84.02%, and 83.63%, respectively, verifying their individual importance in model performance.

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TABLE V

ABLATION STUDY OF THE IMPACT OF DIFFERENT MODULES ON BASELINE PERFORMANCE ON THE ISPRS POTSDAM DATASET

| Models | DSAE | FPAA | PGFRM | OA (%) | mF1 (%) | mIoU (%) |
|---|---|---|---|---|---|---|
| Baseline (Addition Fusion) | × | × | × | 84.72 | 82.25 | 81.33 |
| Baseline + DSAE (Addition Fusion) | ✓ | × | × | 86.43 | 85.24 | 84.76 |
| Baseline + FPAA (Addition Fusion) | × | ✓ | × | 86.15 | 87.01 | 85.31 |
| Baseline + PGFRM | × | × | ✓ | 87.45 | 86.88 | 84.52 |
| GLFRNet | ✓ | ✓ | ✓ | **91.93** | **92.83** | **87.03** |

TABLE VI

ABLATION STUDY OF THE IMPACT OF DIFFERENT MODULES ON BASELINE PERFORMANCE ON THE ISPRS POTSDAM DATASET. FLOPS AND LATENCY ARE SHOWN AS RELATIVE INCREASES VERSUS BASELINE (1.0×) FOR 512 × 512 INPUT

| Models | DSAE | FPAA | PGFRM | OA (%) | mF1 (%) | mIoU (%) | FLOPs (G, rel.) | Latency (ms/img, rel.) |
|---|---|---|---|---|---|---|---|---|
| Baseline | × | × | × | 84.72 | 82.25 | 81.33 | 1.0× | 1.0× |
| Baseline + DSAE | ✓ | × | × | 86.43 | 85.24 | 84.76 | +12% | +9% |
| Baseline + FPAA | × | ✓ | × | 86.15 | 87.01 | 85.31 | +19% | +13% |
| Baseline + PGFRM | × | × | ✓ | 87.45 | 86.88 | 84.52 | +27% | +15% |
| GLFRNet | ✓ | ✓ | ✓ | **91.93** | **92.83** | **87.03** | +37% | +18% |

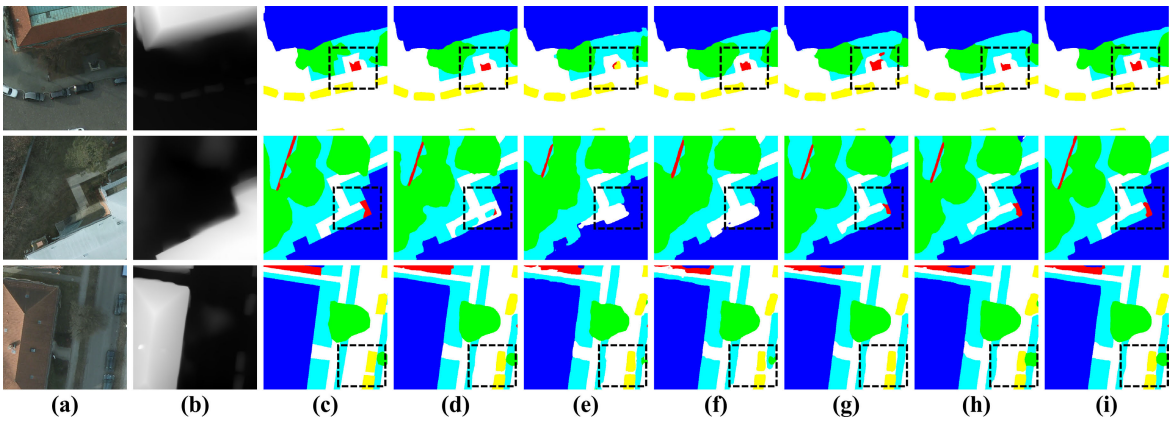



Fig. 10. Comparison of various feature extraction module combinations on the ISPRS Vaihingen test set. (a) Optical. (b) DSM. (c) GT. (d) ResNet50 (Optical) + ResNet50 (DSM). (e) Parameters-shared ResNet50. (f) GBC-Mamba (Optical) + GBC-Mamba (DSM). (g) Parameters-shared GBC-Mamba. (h) ResNet50 (Optical) + GBC-Mamba (DSM). (i) GLFRNet (Ours).

TABLE VII

ABLATION STUDY OF VARIOUS FEATURE EXTRACTION MODULE COMBINATIONS ON THE ISPRS POTSDAM DATASET

| Model | DSAE | FPAA | PGFRM | OA (%) | mF1 (%) | mIoU (%) |
|---|---|---|---|---|---|---|
| ResNet50 (Optical) + ResNet50 (DSM) | × | ✓ | ✓ | 90.83 | 90.27 | 85.74 |
| Parameters-shared ResNet50 | × | ✓ | ✓ | 89.50 | 90.49 | 86.27 |
| GBC-Mamba (Optical) + GBC-Mamba (DSM) | × | ✓ | ✓ | 91.22 | 91.73 | 86.74 |
| Parameters-shared GBC-Mamba | × | ✓ | ✓ | 89.73 | 92.07 | 85.41 |
| ResNet50 (Optical) + GBC-Mamba (DSM) | ✓ | ✓ | ✓ | 90.57 | 91.52 | 85.37 |
| GLFRNet | ✓ | ✓ | ✓ | **91.93** | **92.83** | **87.03** |

From the perspective of computational efficiency, each module introduces a moderate and controllable increase in computational cost. In particular, the integration of PGFRM, which leverages FFT-based frequency-domain processing, results in approximately a 15% to 20% increase in FLOPs and inference latency compared to the version without PGFRM (as evaluated with $512 \times 512$ inputs on a representative GPU). However, this additional overhead is justified by the significant improvement in segmentation accuracy. Visual examples in Fig. 9 confirm that each module addresses specific failure modes, and their combination ensures optimal segmentation detail with a favorable accuracy-efficiency tradeoff for practical applications.

*2) Effect of Feature Extraction Backbone:* Table VII presents a systematic comparison of different feature extraction strategies. Using conventional ResNet50 as the backbone for both modalities yields lower OA and mIoU, indicating that shallow CNN features are insufficient for capturing structural details in high-resolution RSI. In contrast, replacing ResNet50 with the DSAE backbone, which introduces state-space modeling, significantly improves global context understanding and segmentation accuracy; the GBC-Mamba (Optical) + GBC-Mamba (DSM) configuration achieves notable gains across OA, m$F1$, and mIoU. Parameter sharing between modalities reduces model complexity but limits the complementary

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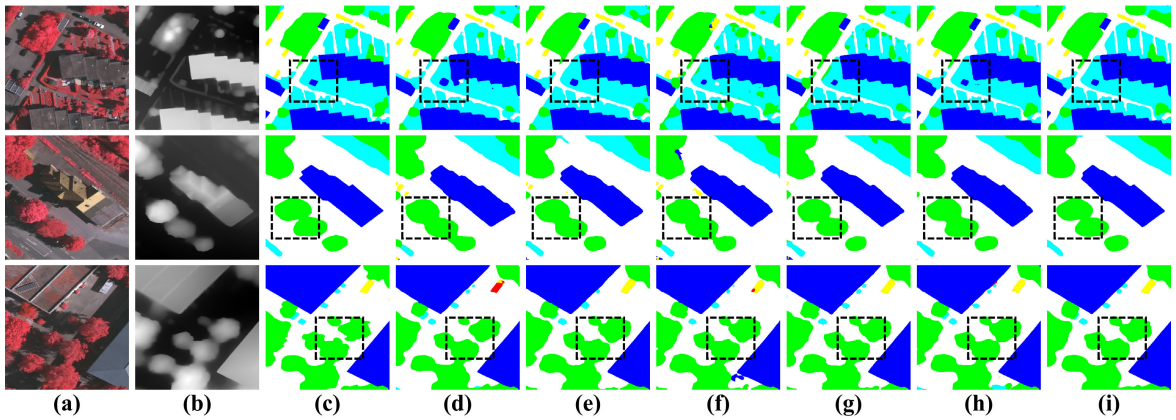



Fig. 11. Comparison of different feature fusion strategies on the ISPRS Vaihingen test set. (a) Optical. (b) DSM. (c) GT. (d) GLFRNet w/o PGFRM + Addition Fusion. (e) GLFRNet w/o PGFRM + Concat Conv Fusion. (f) GLFRNet w/o PGFRM + MBF. (g) GLFRNet w/o PGFRM + FVT. (h) GLFRNet w/o PGFRM + GBF. (i) GLFRNet (Ours).

TABLE VIII
ABLATION STUDY OF DIFFERENT FEATURE FUSION STRATEGIES ON THE ISPRS VAIHINGEN DATASET

| Model | DSAE | FPAA | PGFRM | OA (%) | mF1 (%) | mIoU (%) |
|---|---|---|---|---|---|---|
| GLFRNet+ w/o PGFRM+ Addition Fusion | ✓ | ✓ | × | 91.26 | 90.22 | 83.63 |
| GLFRNet+ w/o PGFRM+ Concat Conv Fusion | ✓ | ✓ | × | 91.69 | 91.46 | 83.41 |
| GLFRNet+ w/o PGFRM+ MBF | ✓ | ✓ | × | 92.34 | 90.17 | 84.20 |
| GLFRNet+ w/o PGFRM+ FVT | ✓ | ✓ | × | 92.02 | **92.50** | 83.74 |
| GLFRNet+ w/o PGFRM+ FFM | ✓ | ✓ | × | 92.46 | 92.25 | 84.47 |
| GLFRNet | ✓ | ✓ | ✓ | **93.27** | 92.33 | **85.35** |

representation of optical and DSM features, resulting in slight performance degradation. These results suggest that independent modeling is essential, as each modality exhibits distinctive semantic distributions at higher levels. Qualitative results in Fig. 10 further support these findings. GLFRNet demonstrates more precise boundary prediction and class consistency, particularly at challenging object boundaries such as building-road and vegetation-tree transitions. The DSAE facilitates effective cross-modal semantic alignment, while FPAA enhances the fusion of HF and LF features, suppressing modality noise and boundary ambiguity. Consequently, GLFRNet achieves the highest mIoU (87.03%) and exhibits superior structural continuity and classification robustness, confirming the key contributions of each module.

*3) Effect of Feature Fusion Strategies:* As shown in Table VIII, different feature fusion mechanisms produce notable performance variations on the ISPRS Vaihingen dataset. Basic fusion methods, such as simple addition and concatenation-convolution, yield lower OA and mIoU, reflecting their limited ability to explicitly model cross-modal semantic dependencies and spatial relationships. More advanced strategies, including Mamba-based fusion (MBF) and feature ViT (FVT), improve global information aggregation and channel selectivity, with FVT particularly enhancing structural priors and HF detail extraction in the frequency domain. However, these approaches still suffer from insufficient feature alignment and modal redundancy in complex scenarios. The introduction of PGFRM leads to the highest performance, with OA, m$F1$, and mIoU reaching 93.27%, 92.33%, and 85.35%, respectively, significantly outperforming all other configurations. This demonstrates the effectiveness of PGFRM in progressive cross-modal reconstruction and salient information guidance. Qualitative results in Fig. 11 further illustrate these differences. Models using simple fusion exhibit boundary errors and object adhesion, especially at building and road interfaces. Gate-based fusion (GBF) shows moderate improvements in structure preservation and class separation. In contrast, the complete GLFRNet accurately recovers object boundaries, reduces class confusion, and maintains clear segmentation between vehicles and the background. These results confirm that PGFRM not only enhances quantitative metrics but also substantially improves spatial consistency.

TABLE IX
QUANTITATIVE COMPARISON OF DIFFERENT ENCODERS ON THE ISPRS VAIHINGEN DATASET

| Encoders | OA(%) | mF1(%) | mIoU(%) |
|---|---|---|---|
| CNN | 91.12 | 88.91 | 82.15 |
| Transformer | 91.08 | 89.83 | 83.47 |
| CNN-Transformer | 91.71 | 90.42 | 84.35 |
| CNN-Mamba | 92.16 | 91.02 | 84.26 |
| DSAE | 93.27 | 92.33 | 85.35 |

*4) Comparison With Existing Encoders:* To further dissect the impact of encoder architecture on multimodal segmentation performance, we conduct a comprehensive comparison among several widely used encoders, including CNN, transformer, hybrid CNN-transformer, CNN-Mamba, and our proposed DSAE. The quantitative results are presented in Table IX, and qualitative comparisons are visualized in Fig. 12.

Compared to traditional CNN and pure transformer encoders, which suffer from limited modeling of long-range dependencies or insufficient spatial inductive bias, our DSAE achieves remarkable improvements, reaching 93.27% in OA, 92.33% in m$F1$, and 85.35% in mIoU. CNN-Mamba

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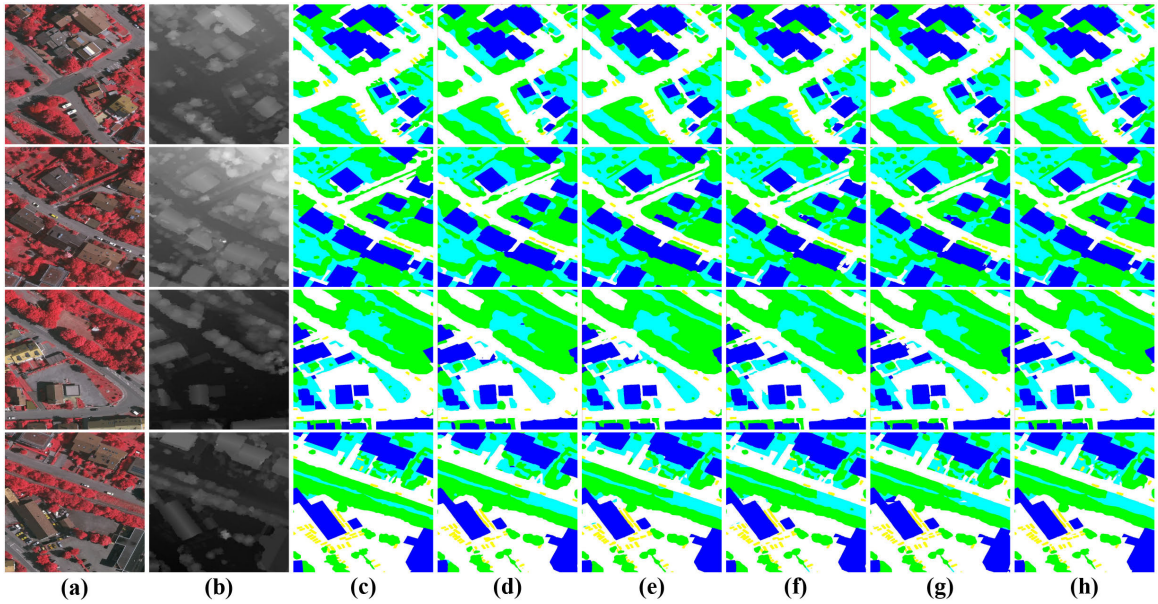



Fig. 12. Quantitative comparison of different encoder architectures on the ISPRS Vaihingen dataset. (a) Optical. (b) DSM. (c) GT. (d) CNN. (e) Transformer. (f) CNN-transformer. (g) CNN-Mamba. (h) DSAE.

TABLE X

QUANTITATIVE COMPARISON OF DIFFERENT ALIGNMENT METHODS ON THE ISPRS POTSDAM DATASET

| Category | OA(%) | mF1(%) | mIoU(%) |
|---|---|---|---|
| Addition Fusion | 90.72 | 90.02 | 85.33 |
| Attention Fusion | 91.21 | 91.16 | 86.02 |
| MLP projection | 90.99 | 90.74 | 85.63 |
| Vision Adapter | 91.05 | 91.01 | 85.82 |
| FPAA | 91.93 | 92.83 | 87.03 |

TABLE XI

SENSITIVITY ANALYSIS OF THE HYPERPARAMETER $\lambda_{\text{align}}$ ON ISPRS POTSDAM VALIDATION. THE BASELINE ($\lambda_{\text{align}} = 0$) REMOVES THE ALIGNMENT LOSS

| $\lambda_{\text{align}}$ | 0 | 0.1 | 0.5 | 1.0 | 2.0 |
|---|---|---|---|---|---|
| OA (%) | 89.87 | 91.23 | **91.93** | 91.62 | 90.95 |
| mF1 (%) | 90.47 | 92.13 | **92.83** | 92.20 | 91.11 |
| mIoU (%) | 85.12 | 86.45 | **87.03** | 86.74 | 85.43 |

and hybrid CNN-transformer encoders achieve moderate gains by introducing sequential modeling or combining convolution with attention, but still lag behind DSAE by over 1% in mIoU. Visual inspection further reveals that DSAE provides significantly sharper semantic boundaries, reduced misclassification of complex regions (e.g., building-road junctions), and enhanced category consistency. While CNN and transformer baselines frequently produce fragmented or noisy segmentation, especially at object edges and fine structures, DSAE demonstrates superior global context integration and cross-modal alignment, evidenced by much cleaner delineation of roads, buildings, and vegetation in challenging areas. These results clearly indicate that the explicit state-space modeling and cross-modality alignment bias introduced by DSAE are critical for achieving fine-grained, reliable predictions in aerial urban scenes.

*5) Comparison With Common Projection Alignment:* To comprehensively evaluate the effectiveness of our proposed FPAA fusion module, we compare it with several widely adopted cross-modal projection and alignment strategies, including element-wise addition fusion, channel attention fusion, MLP-based projection alignment, and vision adapters. Quantitative results on the ISPRS Potsdam dataset are summarized in Table X.

Our FPAA achieves the highest performance across all metrics, with 91.93% OA, 92.83% m$F1$, and 87.03% mIoU, consistently outperforming all mainstream baselines. Notably, compared with conventional addition fusion and MLP projection, which provide only basic or linear feature interaction, FPAA delivers an absolute gain of 1.70% and 1.40% in mIoU, respectively. Attention fusion and vision adapters, which model interchannel or lightweight adaptive alignment, also fall short of FPAA, illustrating that frequency-aware feature modulation enables more discriminative and complementary cross-modal integration. Visual comparisons in Fig. 13 further substantiate these advantages. While existing fusion approaches tend to produce blurred boundaries or inconsistent predictions in complex urban regions (e.g., building edges, small structures, and cluttered vegetation), FPAA effectively suppresses noise and preserves delicate object contours. In particular, the predictions from FPAA exhibit sharper transitions and more precise delineation of fine-grained categories, faithfully aligning with ground truth segmentation. These results confirm that explicitly modeling frequency-domain interactions is crucial for robust and detail-preserving multimodal fusion in semantic segmentation.

*6) Sensitivity Analysis of the Alignment Loss Weight:* To assess the influence of the alignment loss on our training regime, we conduct a sensitivity analysis on the hyperparameter $\lambda_{\text{align}}$, which controls the relative weight of the cross-modal alignment loss in the total optimization objective. Table XI summarizes the performance variations on

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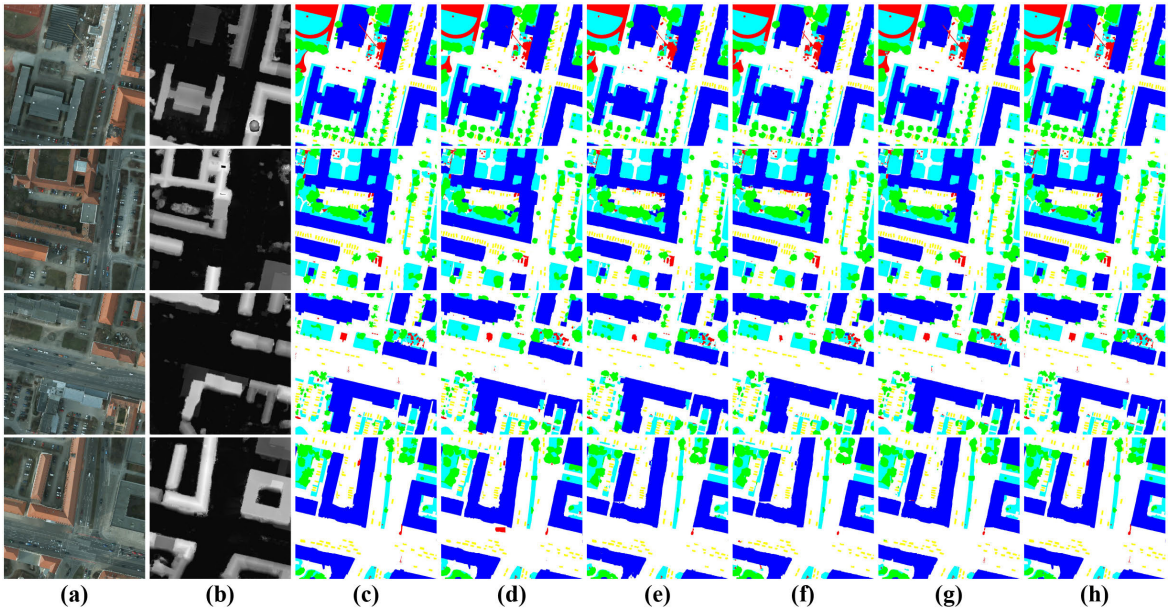



Fig. 13. Quantitative comparison of different fusion methods on the ISPRS Potsdam dataset. (a) Optical. (b) DSM. (c) GT. (d) Addition fusion. (e) Attention fusion. (f) MLP projection. (g) Vision adapter. (h) FPAA.

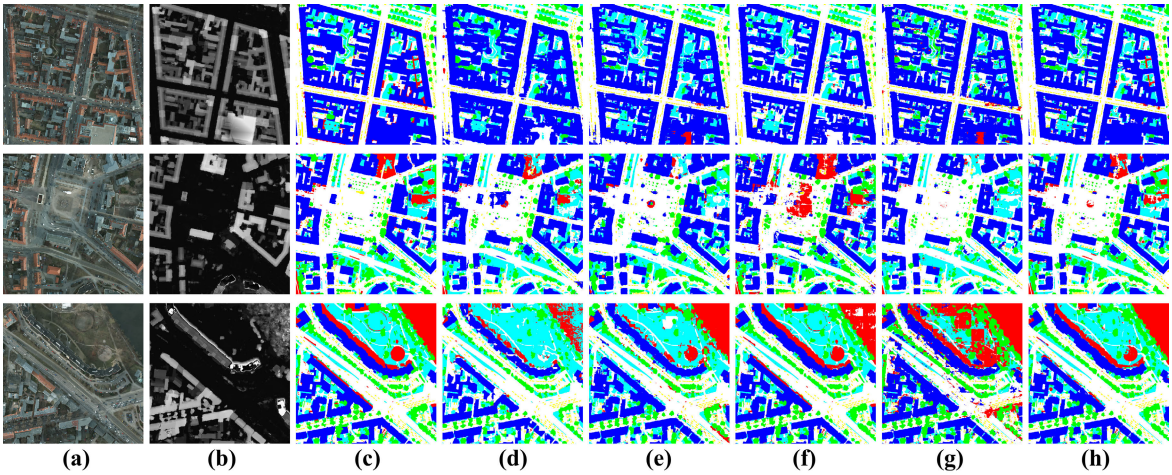



Fig. 14. Visual comparison of segmentation results for models trained on Vaihingen and tested on Potsdam. (a) Optical. (b) DSM. (c) GT. (d) TransUNet. (e) FuseNet. (f) SA-GATE. (g) MultiSenseSeg. (h) GLFRNet (Ours).

the ISPRS Potsdam validation set with $\lambda_{\text{align}}$ ranging from 0 (alignment loss disabled) to 2.0. The results show that incorporating the alignment loss ($\lambda_{\text{align}} > 0$) consistently improves OA and segmentation quality compared to the baseline without alignment guidance ($\lambda_{\text{align}} = 0$). Notably, the best performance is achieved when $\lambda_{\text{align}} = 0.5$, with mIoU reaching 87.03%.

Increasing $\lambda_{\text{align}}$ further beyond this point leads to a gradual decline in accuracy, suggesting that excessive emphasis on alignment may disrupt the optimal balance with the core segmentation objective. These trends indicate that the model benefits from moderate alignment regularization, which reinforces both spatial and frequency-domain consistency between modalities while maintaining semantic discriminability. Meanwhile, over-penalizing misalignment may overly constrain the feature learning and impact the main task. In practice, we select $\lambda_{\text{align}}$ within the range [0.1, 1.0] to ensure balanced segmentation performance.

### G. Domain Generalization

To evaluate the generalization capability of multimodal segmentation models under domain shift, we conduct a cross-domain experiment where all models are trained on the ISPRS Vaihingen dataset and directly tested on the ISPRS Potsdam dataset without any fine-tuning. As shown in Table XII, all methods experience a notable performance drop compared to in-domain testing, highlighting the significant challenges posed by cross-dataset generalization in remote sensing. Nevertheless, GLFRNet achieves the highest accuracy among the compared methods, with an OA of 63.64%, m$F1$ of 58.62%, and mIoU of 56.50%, outperforming other advanced multimodal baselines. Notably, GLFRNet demonstrates consistent improvements across most semantic categories, particularly for building, tree, and car classes, indicating its superior ability to leverage complementary information from optical and DSM modalities. Qualitative results in Fig. 14 further confirm that GLFRNet delivers more robust and spatially

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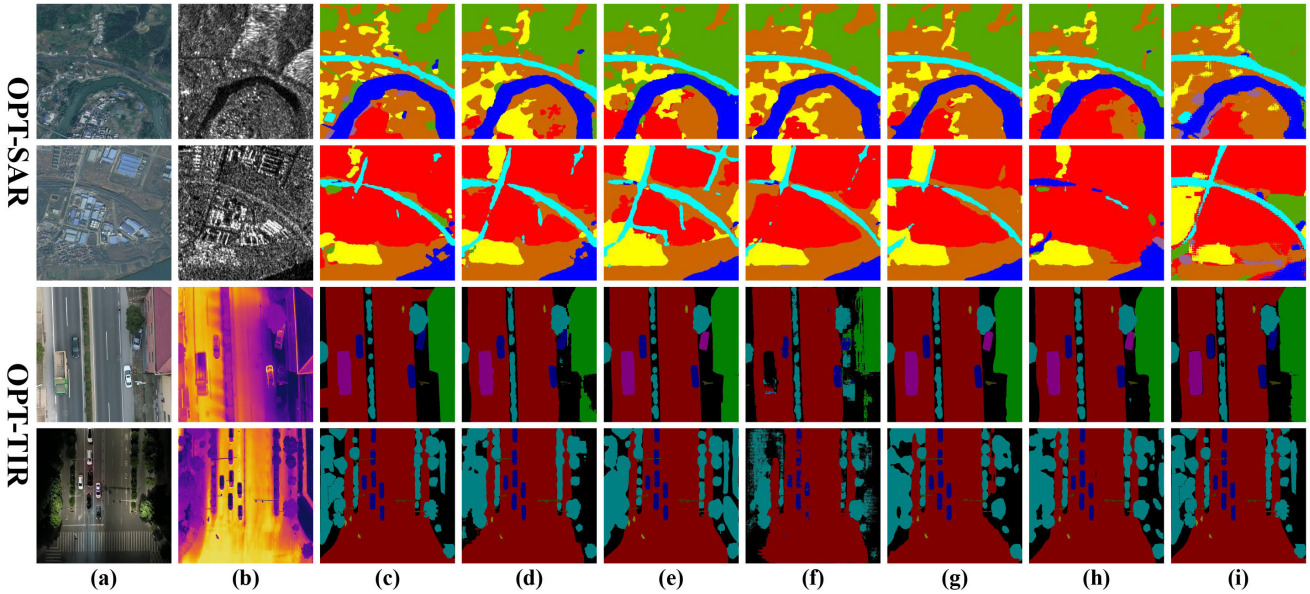



Fig. 15. Qualitative segmentation comparisons on OPT-SAR and OPT-TIR datasets. (a) Optical. (b) SAR/TIR. (c) GT. (d) TransUNet. (e) FuseNet. (f) SA-GATE. (g) MultiSenseSeg. (h) FTransUNet. (i) GLFRNet (Ours).

TABLE XII

QUANTITATIVE COMPARISON OF DOMAIN GENERALIZATION PERFORMANCE BY TRAINING ON THE ISPRS VAIHINGEN DATASET AND TESTING ON THE ISPRS POTSDAM DATASET

| Type | Method | Bui. | Tre. | Low. | Car | Imp. | Total | mF1(%) | mIoU(%) |
|---|---|---|---|---|---|---|---|---|---|
| Multimodal | TransUNet | 51.30 | 48.02 | 44.95 | 52.60 | 50.85 | 46.64 | 53.72 | 40.79 |
| | FuseNet | 58.64 | 55.01 | 53.61 | 50.69 | 59.15 | 53.19 | 42.31 | 44.28 |
| | SA-GATE | 55.88 | 61.83 | 47.73 | 55.92 | 46.82 | 47.62 | 47.27 | 52.74 |
| | MultiSenseSeg | 48.02 | 56.78 | 55.08 | 47.46 | 49.62 | 61.42 | 52.04 | 45.31 |
| | FTransUNet | 52.87 | 50.38 | **61.38** | 54.33 | 56.41 | 50.16 | 46.85 | 50.73 |
| | GLFRNet | **61.98** | **64.98** | 58.37 | **64.43** | **61.57** | **63.64** | **58.62** | **56.50** |

consistent segmentation, with clearer object boundaries and reduced misclassification in complex scenes. Despite these advances, the overall performance remains suboptimal for practical applications, suggesting that domain shift remains a fundamental challenge and motivating further research into explicit domain adaptation and generalization strategies.

### *H. Cross-Modal Analysis*

To further assess the generality of our approach, we conducted extensive experiments on two challenging and highly disparate cross-modal remote sensing datasets: WHU-OPT-SAR and RS-RGB-T. WHU-OPT-SAR [56] comprises co-registered optical, near-infrared, and SAR images, presenting significant modality disparity and variation in imaging physics. The RS-RGB-T [57] dataset includes RGB and thermal infrared (TIR) urban scene pairs captured under diverse illumination and weather conditions. As detailed in Table XIII, GLFRNet achieves state-of-the-art performance across both the OPT-SAR and OPT-TIR benchmarks.

On the challenging WHU-OPT-SAR dataset, our model obtains an OA of 87.02%, m$F1$ of 78.65%, and mIoU of 65.17%. These results represent absolute gains of +4.91%, +4.34%, and +6.26% in OA, m$F1$, and mIoU, respectively, compared to the strongest competing baseline (FTransUNet). On the RS-RGB-T benchmark (OPT-TIR), GLFRNet sets a new performance record with 88.52% OA, 87.15% m$F1$, and 79.35% mIoU, consistently outperforming recent transformer-based fusion models by a considerable margin. These quantitative improvements are further corroborated by the qualitative segmentation results in Fig. 15. Specifically, GLFRNet generates more coherent and sharper object boundaries, accurately delineates small and elongated structures (such as

TABLE XIII

COMPARISON WITH STATE-OF-THE-ART MULTIMODAL SEGMENTATION METHODS ON WHU-OPT-SAR AND RS-RGB-T DATASETS

| Dataset Type | Method | OA (%) | mF1 (%) | mIoU (%) |
|---|---|---|---|---|
| OPT-SAR | TransUNet | 78.92 | 70.57 | 55.47 |
| | FuseNet | 81.17 | 72.30 | 57.22 |
| | SA-GATE | 81.15 | 73.06 | 57.62 |
| | MultiSenseSeg | 82.39 | 73.41 | 58.91 |
| | FTransUNet | 82.11 | 74.31 | 58.64 |
| | GLFRNet | **87.02** | **78.65** | **65.17** |
| OPT-TIR | TransUNet | 83.73 | 79.45 | 72.61 |
| | FuseNet | 85.92 | 80.07 | 74.33 |
| | SA-GATE | 85.32 | 79.33 | 73.67 |
| | MultiSenseSeg | 85.37 | 80.19 | 74.17 |
| | FTransUNet | 86.47 | 81.67 | 76.01 |
| | GLFRNet | **88.52** | **87.15** | **79.35** |

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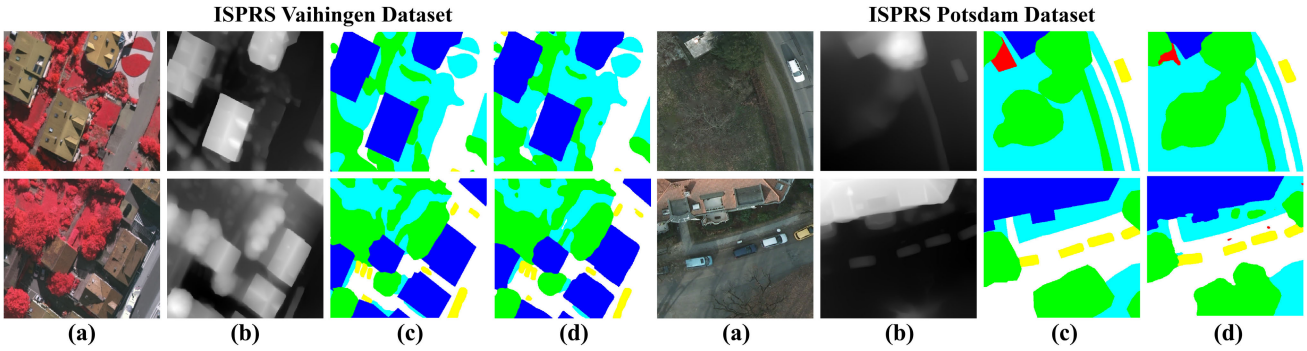



Fig. 16. Visualization of failure case results. (a) Optical. (b) DSM. (c) GT. (d) GLFRNet (Ours).

roads and watercourses), and maintains semantic integrity in complex scenes with substantial cross-modal variation. Different from prior methods, which often fail to resolve detail or are challenged by strong sensor domain gaps, our approach demonstrates marked robustness to heterogeneous appearance, illumination changes, and sensor artifacts. These results confirm the strong cross-modal fusion capability of GLFRNet, making it suitable for a wide range of practical applications involving disparate sensor modalities.

### I. Failure Case Analysis

As illustrated by the qualitative results in Fig. 16, despite the strong overall segmentation performance of GLFRNet, several typical failure cases remain. In shadowed areas or between classes with similar spectral signatures (e.g., buildings and roads with similar roofing), misclassification often arises, mainly due to the limitations of the optical encoder and incomplete cross-modal alignment in FPAA. DSM noise or occlusion (from trees or sensor artifacts) can distort height cues and boundaries, reflecting limited robustness in both FPAA and frequency-domain compensation in PGFRM. Slight misregistration between optical and DSM inputs leads to fragmented contours, indicating that explicit spatial alignment is still insufficient. Furthermore, small or low-contrast objects (such as vehicles or narrow road markings) may be omitted or blurred due to limited saliency modeling and potential information loss during fusion. Most of these errors occur at challenging boundaries or low-confidence regions, highlighting the need for further improvements in modality alignment, noise handling, and small-object enhancement.

## V. Conclusion

In this article, we introduced GLFRNet, a unified multimodal semantic segmentation framework that explicitly addresses the challenges of global–local feature integration, cross-modal alignment, and frequency-domain information preservation. Technically, GLFRNet incorporates a DSAE that combines convolutional and state space modeling for comprehensive local–global feature extraction; a global–local FPAA that enforces semantic and structural consistency across modalities through global projection and pixel-wise alignment; and a PGFRM that leverages frequency decomposition and phase-guided fusion to recover complementary information from optical and DSM data. Extensive experiments on ISPRS Vaihingen and Potsdam datasets demonstrate that GLFRNet achieves superior performance in terms of mIoU, OA, and m$F1$, particularly excelling in complex structural regions. Ablation studies validate the contribution of each component. Future work will explore lightweight implementations and advanced adaptation strategies to enhance scalability and practical deployment in diverse remote sensing applications.

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